

FRACTIONAL AFFINE AREA MEASURES

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ABSTRACT. Inspired by recent progress in integral geometry, (fractional) affine area measures are introduced. They can be viewed as differentials of (fractional) integral affine surface area ($\text{SL}(n)$ -invariants). Existence and uniqueness results for the Minkowski problem of these measures are provided. Connections to a long-standing conjecture in affine integral geometry are shown.

1. INTRODUCTION

The classical isoperimetric inequality that compares the volume and the surface area of a convex body (compact, convex subsets of $\mathbb{R}^n$ with nonempty interior):

$$n\omega_n^{\frac{1}{n}}V(K)^{\frac{n-1}{n}} \leq S(K) \quad \text{with equality if and only if } K \text{ is a ball,} \quad (1.1)$$

is arguably the most fundamental and one of the oldest results in mathematics. Here, we use ω_n for the volume of the n -dimensional unit ball. It dates back to at least the ancient Greeks, while the first rigorous proof of the inequality, where symmetrization techniques played an essential role, appeared in the 19th century.

From an integral geometry viewpoint, *Cauchy's integral formula* tells us that surface area can be represented, up to a constant, as the average of areas of projections:

$$S(K) = \frac{1}{\omega_{n-1}} \int_{S^{n-1}} \text{vol}_{n-1}(K|u^\perp) du.$$

Here $K|u^\perp$ denotes the image of the orthogonal projection of K onto $u^\perp$. While volume is *affine invariant* in the sense that $V(\phi K + x_0) = V(K)$ for every $\phi \in \text{SL}(n)$ and $x_0 \in \mathbb{R}^n$, the surface area $S(K)$ is not. Instead, up to a constant, by taking the L_{-n} mean, one obtains the *integral affine surface area*

$$\Phi(K) = \left(\frac{1}{n} \int_{S^{n-1}} \text{vol}_{n-1}(K|u^\perp)^{-n} du \right)^{-1/n},$$

which is affine invariant. The celebrated *Petty projection inequality* (see e.g., [63]), which is stronger than the classical (1.1), states:

$$n\omega_n^{\frac{1}{n}}V(K)^{\frac{n-1}{n}} \leq \frac{n\omega_n^{(n+1)/n}}{\omega_{n-1}} \Phi(K) \leq S(K).$$

Note here, the first inequality (*i.e.*, the Petty projection inequality) is affine invariant in the sense that all terms involved in it are invariant under the map $K \mapsto \phi K + x_0$ for every $\phi \in \text{SL}(n)$

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and $x_0 \in \mathbb{R}^n$. Consequently, equality holds if and only if K is an ellipsoid. The Petty projection inequality is the geometric core of the *affine Zhang-Sobolev inequality* [69] that strengthens the classical (non-affine) L_1 Sobolev inequality. It is worth mentioning that the integral affine surface area is one of the affine quermassintegrals introduced in 1984 by Lutwak, who conjectured [48] that, similar to the Petty projection inequality, among all convex bodies of equal volume, affine quermassintegrals are minimized precisely at ellipsoids. This conjecture was recently completely resolved by Milman-Yehudayoff [55].

A “differential” treatment of the geometric invariants involved in the isoperimetric inequality, particularly the volume, appeared when Minkowski asked the following question at the turn of the previous century: Consider unit vectors $v_1, \dots, v_N$ and positive reals $a_1, \dots, a_N$, is there a polytope whose facet normals are given by v_i while its facet areas are given by a_i ? Aleksandrov [1] in 1939 revealed that the data $(v_i \text{ and } a_i)$ prescribed in Minkowski’s problem can be defined for every convex body K in the form of a finite Borel measure on S^{n-1} , now known as the *surface area measure* $S_{n-1}(K, \cdot)$, see (2.2), and that it is the differential of the volume operator:

$$\left. \frac{d}{dt} \right|_{t=0} V([h_0 + tf]) = \int_{S^{n-1}} f(v) dS_{n-1}([h_0], v). \quad (1.2)$$

Here h_0, f are continuous functions on S^{n-1} and

$$[h_0 + tf] = \{x \in \mathbb{R}^n : x \cdot v \leq h_0(v) + tf(v), \text{ for all } v \in S^{n-1}\}$$

denotes the *Wulff shape* generated by $h_0 + tf$. Aleksandrov’s variational formula holds as long as $[h_0]$ defines a convex body, without *any* additional regularity assumptions. Remarkably, the derivative holds even for $t \rightarrow 0^-$, considering especially that $h_0 + tf \mapsto [h_0 + tf]$ is, *loosely* speaking, a highly *non-smooth* operation. Aleksandrov’s variational formula brought the technique of *calculus of variation* to the classical Minkowski problem.

The classical Minkowski problem. Given a finite Borel measure μ on S^{n-1} , find the necessary and sufficient conditions on μ so that $\mu = S_{n-1}(K, \cdot)$ for some convex body K . If such a K exists, is it unique? How much regularity does K have if μ satisfies further regularity assumptions?

The classical Minkowski problem is known as the problem of prescribing Gauss curvature in differential geometry and takes the form of the Monge-Ampère equation in PDE. It was studied throughout the entire last century, as demonstrated by the works of Minkowski [56], Aleksandrov [1], Cheng-Yau [18], Pogorelov [57], and Caffarelli [11, 12, 13].

The other geometric invariant that appears in the isoperimetric inequality, surface area, turned out to be much more *ill-behaved* under perturbations of the body. We will write $K + L = \{x + y : x \in K, y \in L\}$ to be the Minkowski sum of two convex bodies K and L . An application of the Steiner expansion formula shows

$$\left. \frac{d}{dt} \right|_{t=0^+} S(K + tL) = (n-1) \int_{S^{n-1}} h_L(v) dS_{n-2}(K, \cdot), \quad (1.3)$$

where h_L is the support function (see (2.1)) of L and $S_{n-2}(K, \cdot)$ is the $(n-2)$ -th area measure of K introduced by Aleksandrov, Fenchel, and Jessen in the 1930s. With additional smoothness assumptions, the density of $S_{n-2}(K, \cdot)$ involves the elementary symmetric polynomials of principal radii of curvatures. The formula (1.3) suggests the differential of surface area *should* lead to the area measure $S_{n-2}(K, \cdot)$. However, note that the derivative in (1.3) is only *one-sided*, and it only works with convexity assumptions on K and L . Compare this to (1.2) which holds with much

greater robustness. In fact, volume and surface area are special cases of a bigger family of integral geometric invariants, called *quermassintegrals*¹. It is sadly the truth that volume is the “unicorn” in the family of quermassintegrals, in that it is the only one that permits a variational formula as robust as (1.2). Because area measures, such as $S_{n-2}(K, \cdot)$, are associated with elementary symmetric polynomials of principal radii of curvatures, their Minkowski-type characterization problem (for example, solving the equation $\mu = S_{n-2}(K, \cdot)$) is of significant interest. These are known as *Christoffel-Minkowski problems*. The lack of *robust* variational formulas makes the method of *calculus of variation* much more difficult to apply, which is one of the primary reasons that these problems are still long-standing, although significant contributions have been made using PDE techniques. See, for example, Guan-Guan [27], Guan-Li-Li [28], Guan-Ma [29], Guan-Ma-Zhou [30]. See, also, Firey and Berg (Section 8.3.2 in [63]), Grinberg-Zhang [26], Goodey-Yaskin-Yaskina [25], Schneider [62] for the solution of the Christoffel problem.

Nonetheless, in a landmark work [51] by Lutwak-Xi-Yang-Zhang, the authors paved an alternative route to $S_{n-2}(K, \cdot)$ by investigating geometric measures in integral geometry. For $q > 0$, the *chord integral* $I_q(K)$ is the L_q mean of the length of intersection of the body K with a random line in $\mathbb{R}^n$:

$$I_q(K) = \int_{\mathcal{L}^n} |K \cap \ell|^q d\ell$$

where the integration is with respect to the (appropriately normalized) Haar measure on the affine Grassmannian $\mathcal{L}^n$ of lines in $\mathbb{R}^n$. Lutwak-Xi-Yang-Zhang established a family of *robust* variation formulas for *chord integrals* $I_q(K)$, leading to *chord measures* $F_q(K, \cdot)$. As $q \rightarrow 0$, the chord integral $I_q(K)$ converges to the surface area $S(K)$. It is surprising that, as $q \rightarrow 0$, the chord measure $F_q(K, \cdot)$ also converges to $S_{n-2}(K, \cdot)$. If uniform (in q) estimates can be obtained for the measure equation $\mu = F_q(K, \cdot)$, then a limiting argument will lead to a solution to the Christoffel-Minkowski problem for $S_{n-2}(K, \cdot)$. We remind the readers that a *robust* variational formula for surface area does *not* exist. Naturally, as $q \rightarrow 0^+$, singularities arise when proving the variational formula for $I_q(K)$, especially when $q \in (0, 1)$ and delicate handling of these singularities is required.

It is natural to ask: what is the *affine area measure* $S_{n-2}^{(a)}(K, \cdot)$ related to the integral affine surface area, such that

$$\left. \frac{d}{dt} \right|_{t=0^+} \Phi(K + tL) = (n-1) \int_{S^{n-1}} h_L(v) dS_{n-2}^{(a)}(K, \cdot)? \quad (1.4)$$

This task and more importantly, characterizing the derived measure are naturally crucial problems in affine integral geometry. In Section 4, we introduce the affine area measure $S_{n-2}^{(a)}(K, \cdot)$ (Definition 4.3) and establish (1.4). It is interesting to see that classical constructions such as polar projection body and centroid body naturally appear in this definition. Note that the derivative is one-sided. With the additional assumption that K has C^2 boundary and everywhere positive Gauss curvature, we show the validity of (1.4) for the two-sided derivative. What is involved in the affine area measure is a weighted version of the normal curvature where the weight is connected to the projectional areas of K onto lower-dimensional subspaces. It is also worth pointing out that there are three separate (yet equivalent) ways we could have introduced the affine area measure $S_{n-2}^{(a)}(K, \cdot)$. See Remark 4.11. Moreover, we show that if $S_{n-2}^{(a)}(K, \cdot) = S_{n-2}^{(a)}(L, \cdot)$,

¹which up to a constant is equal to the average integral of i -dimensional orthogonal projection area of the convex body. (Volume is the quermassintegral when $i = n$, whereas surface area corresponds to $i = n - 1$.)

then K and L differ from each other by a translation. Unfortunately, given the negative answer we have for the surface area $S(\cdot)$, it seems implausible that a double-sided derivative formula can hold for a generic convex body K without *any* additional regularity assumptions. This makes it extremely challenging to answer a Minkowski-type problem for the affine area measure $S_{n-2}^{(a)}(K, \cdot)$, at least from a calculus of variation point of view. However, an alternative route to approach $S_{n-2}^{(a)}(K, \cdot)$ is possible. We will demonstrate variational formulas for a family of *affine* invariants that contains the integral affine surface area as the limiting case. Moreover, as it will become clear, the limit of the measures appearing in these variational formulas is precisely the affine area measure $S_{n-2}^{(a)}(K, \cdot)$.

Projection body ΠK and difference body DK are natural constructions in affine geometry, see Ludwig [44] and Gardner-Hug-Weil [22]. Coincidentally, the integral affine surface area $\Phi(K)$ can be represented as the volume of the *polar body* of the projection body $\Pi^*K := (\Pi K)^*$ (see (2.5) for its precise definition),

$$\Phi(K) := V(\Pi^*K)^{-\frac{1}{n}}.$$

As revealed by Gardner-Zhang [24], polar projection body and difference body appear as special cases of the *radial q -th mean body* R_qK defined for each $q > -1$. See (2.4) for its precise definition. Intuitively, in each direction u , the length of the 1-dimensional section of R_qK in the direction u captures the L_q mean of the length of K in the direction u measured starting from a random point $x \in K$. It sometimes makes various formulas neater to consider a normalized q -th mean body given by $S_q(K) = [(q+1)V(K)]^{\frac{1}{q}} R_qK$. The (normalized) radial q -th mean body is $\text{SL}(n)$ -equivariant: for $\phi \in \text{SL}(n)$, we have $S_q(\phi K) = \phi S_qK$. This makes its volume $\text{SL}(n)$ -invariant. The difference body corresponds to the $q = \infty$ case, whereas up to a normalizing constant, the polar projection body appears as $q \rightarrow -1^+$. Consequently, the integral affine surface area $\Phi(K)$ can be recovered:

$$\lim_{q \rightarrow -1^+} V(S_qK)^{\frac{q}{n}} = \Phi(K). \quad (1.5)$$

For this reason, we denote the left-side of (1.5) as $\Phi_q(K)$; that is, for $q > -1$,

$$\Phi_q(K) = V(S_qK)^{\frac{q}{n}} = (q+1)V(K)V(R_qK)^{\frac{q}{n}}.$$

Special cases include the integral affine surface area $\Phi_{-1}(K) = \Phi(K)$ in the sense of limit, and when $q = n$, $\Phi_n(K) = (n+1)V(K)^2$. It is natural, especially when $q \in (-1, n)$, to call $\Phi_q(K)$ the *q -th fractional integral affine surface area* of K . As the name suggests, for each $q \geq -1$, the invariant $\Phi_q(K)$ is $\text{SL}(n)$ -invariant and translation invariant; that is, for $\phi \in \text{SL}(n)$ and $x_0 \in \mathbb{R}^n$, we have $\Phi_q(\phi K + x_0) = \Phi_q(K)$.

One of the main results in the current work is to show that it is possible to differentiate $\Phi_q(K)$ for *every* $-1 < q \neq 0$ and that a *robust* variational formula (as powerful as that of Aleksandrov's) can be established.

Theorem 1.1. *Let $0 \neq q > -1$, $\Omega \subset S^{n-1}$ be a compact subset not contained in any closed hemisphere, and $h_0, g \in C(\Omega)$. Suppose that for sufficiently small $\delta > 0$ and $|t| < \delta$, the Wulff shape² $K_t = [h_t]$ where*

$$h_t = h_0 + tg + o(t, \cdot), \text{ on } \Omega$$

²See Section 2 for its definition.

is a convex body in $\mathbb{R}^n$. Here $o(t, \cdot)/t \rightarrow 0$ uniformly on Ω as $t \rightarrow 0$. Then,

$$\left. \frac{d}{dt} \right|_{t=0} \Phi_q(K_t) = 2 \Phi_q^{\frac{q-n}{q}}(K_0) \int_{\Omega} g(v) dF_q^{(a)}(K_0, v).$$

Here, the measure $F_q^{(a)}(K, \cdot)$ is consequently called the q -th *fractional affine area measure* of K . See Definition 3.1 for its precise definition—we are forced to delay its presentation until then because many notations to be introduced in Section 2 are needed for the definition. We remark that $F_q^{(a)}(K, \cdot)$ is a *finite* Borel measure on S^{n-1} , which is *not* straightforward when $q \in (-1, 0)$. See Corollary 3.5. In fact, a careful handling of singularities arising in its definition is of utmost importance. We point out that Theorem 3.7 contains a subtler result that allows one to differentiate a modified version of the chord integral. Consequently, the treatment adopted here can be adapted to yield a more direct and shorter proof of Lutwak-Xi-Yang-Zhang [51, Theorem 5.5]. It is also worth mentioning that when $q = 0$, Theorem 3.7 recovers Aleksandrov's variational formula (1.2).

Fractional affine area measures inherit a few properties from Φ_q : they are translation invariant $F_q^{(a)}(K + x_0, \cdot) = F_q^{(a)}(K, \cdot)$ and the centroid of $F_q^{(a)}(K, \cdot)$ is at the origin. The proofs for these properties (and a few more) can be found in Section 3.

As will become clear in Section 4, as $q \rightarrow -1^+$, with additional regularity assumptions, these fractional affine area measures converge *set-wise* to the affine area measure $S_{n-2}^{(a)}(K, \cdot)$.

The following problem is natural.

The Minkowski problem for fractional affine surface area measure. Let $0 \neq q > 1$ and μ be a finite Borel measure on S^{n-1} . What are the necessary and sufficient condition(s) on μ so that there exists a convex body K in $\mathbb{R}^n$ such that

$$\mu = F_q^{(a)}(K, \cdot)?$$

We present the following solution to the Minkowski problem for $F_q^{(a)}(K, \cdot)$.

Theorem 1.2. *Let $0 \neq q > -1$ and μ be a finite Borel measure on S^{n-1} . There is a convex body K in $\mathbb{R}^n$ such that $\mu = F_q^{(a)}(K, \cdot)$ if and only if μ is not concentrated in any subsphere and*

$$\int_{S^{n-1}} v d\mu(v) = o. \tag{1.6}$$

With additional regularity assumptions, the measure equation $\mu = F_q^{(a)}(K, \cdot)$ reduces to the rather complicated Monge-Ampère equation (5.2) for $p = 1$. We emphasize that even in this special case, equation (5.2) is *nonlocal*; that is, it is impossible to verify a solution if it is only given in a local neighborhood. Previously, this phenomenon has only occurred very recently in the work Lutwak-Xi-Yang-Zhang [51], where they introduced the first geometric measures and Minkowski-type problems (consequently PDEs) in integral geometry.

Following the two pioneering works [49, 50] by Lutwak in the early 1990s, many elements in the Brunn-Minkowski theory, including surface area measure and the classical Minkowski problem, have been extended to their L_p counterparts. This has occupied much of the research in convex geometric analysis and fully nonlinear elliptic PDEs (particularly those of Monge-Ampère type) in the last two decades. See, for example, [2, 5, 19, 32, 34, 38, 46, 45, 52, 53, 64, 65, 67] for a (not even close to exhaustive) list of works in the L_p Brunn-Minkowski theory. In particular, the theory becomes *significantly* harder when $p < 1$. Neither the isoperimetric inequalities nor

the Minkowski problems have been fully addressed. In particular, the *log Minkowski problem* (for the *cone volume measure*, or, the case $p = 0$) has not yet been fully solved. See, for example, Bianchi-Böröczky-Colesanti-Yang [3], Böröczky-Lutwak-Yang-Zhang [8], Chou-Wang [19], Guang-Li-Wang [31], Zhu [71, 72] among many other works. The isoperimetric inequality in the $p = 0$ case is the *log Brunn-Minkowski conjecture* (see, for example, Böröczky-Lutwak-Yang-Zhang [7])—arguably one of the most crucial conjectures in convex geometric analysis in the past decade. The log Brunn-Minkowski conjecture has been verified in dimension two and in various special classes of convex bodies. See, for example, Chen-Huang-Li-Liu [16], Colesanti-Livshyts-Marsiglietti [20], Kolesnikov-Livshyts [41], Kolesnikov-Milman [40], Milman [54], Putterman [58], Saroglou [61]. If proven correct, it is much stronger than the classical Brunn-Minkowski inequality.

Inspired by this, for $p \in \mathbb{R}$ and a convex body K containing the origin in its interior, we define the L_p fractional integral affine surface area of the q -th order by

$$F_{p,q}^{(a)}(K, \cdot) = h_K^{1-p} dF_q^{(a)}(K, \cdot).$$

The L_p Minkowski problem for fractional affine area measure. Let $0 \neq q > -1$, $p \in \mathbb{R}$, and μ be a finite Borel measure on S^{n-1} . What are the necessary and sufficient condition(s) on μ so that there exists a convex body K in $\mathbb{R}^n$ (containing the origin in its interior if $p \neq 1$) such that

$$\mu = F_{p,q}^{(a)}(K, \cdot)?$$

The singular case $p = 0$ is special in this family. In fact, the measure $F_{0,q}^{(a)}(K, \cdot)$ is the only $\text{SL}(n)$ -contravariant one in this family. See Theorem 3.15 and the definition of $\text{SL}(n)$ -contravariant measures immediately preceding it. We prove the following existence theorem in this special case.

Theorem 1.3. *Let $0 \neq q > -1$ and μ be a finite even Borel measure on S^{n-1} . If μ satisfies the strict subspace concentration condition; that is*

$$\frac{\mu(\xi \cap S^{n-1})}{\mu(S^{n-1})} < \frac{\dim \xi}{n}, \quad (1.7)$$

for every proper subspace $\xi \subset \mathbb{R}^n$, then there exists an origin-symmetric convex body K such that

$$\mu = F_{0,q}^{(a)}(K, \cdot).$$

It is worth pointing out that Klartag [39, Corollary 5.3] demonstrated that measures on S^{n-1} satisfying (1.7) are *potentially isotropic*; that is, there exists $T \in \text{SL}(n)$ such that its affine image $T\mu$ is isotropic.³ See (3.11) for the definition of $T\mu$. See also Böröczky-Lutwak-Yang-Zhang [9] for a necessary and sufficient condition for a Borel measure μ on S^{n-1} to be potentially isotropic.

It is of great interest to investigate if a Brunn-Minkowski type inequality or even a log-Brunn-Minkowski type inequality can be formulated for $\Phi_q(K)$ and how they are connected to the uniqueness of solutions to the Minkowski problems for $F_{p,q}^{(a)}(K, \cdot)$.

Another motivation for investigation into $\Phi_q(K)$ and particularly Theorem 1.1 comes from the perspective of sharp affine isoperimetric inequalities. Recently, Haddad-Ludwig [33] resolved a longstanding conjecture, the treatment of which involves heavily Riesz-type integral, the Riesz rearrangement inequality (see, for example, Burchard [10]), as well as fractional perimeter. Their results can be stated as:

³In general, Klartag's result applies to Borel measures on $\mathbb{R}^n$ as long as $\mu(\{o\}) = 0$.

Theorem 1.4 ([33]). *If $-1 < p \leq n \leq q$ with $p \neq q$, then the $GL(n)$ -invariant ratio*

$$\frac{V(R_p K)}{V(R_q K)} \quad (1.8)$$

attains its maximum on the set of convex bodies in $\mathbb{R}^n$ if and only if K is an ellipsoid.

The reverse of Theorem 1.4 (that is, the minimum of the ratio on the set of convex bodies) was previously shown by Gardner-Zhang [24] and the minimum is attained if and only if K is a simplex. Contained as special cases of Haddad-Ludwig [33] and Gardner-Zhang [24] are well-known sharp affine isoperimetric inequalities: the Rogers-Shephard inequality [59, 60], the Petty Projection inequality and its reverse (often called the Zhang projection inequality [68]). Theorem 1.1 makes it possible to treat these sharp affine isoperimetric inequalities and generally the investigation of the ratio (1.8) through its critical points. The Gardner-Zhang inequality [24] and the Haddad-Ludwig inequality [33] imply: ellipsoids and simplices are the critical points of (1.8). In Section 6, we derive measure equations (and in special cases, integral equations) satisfied by critical points of the ratio (1.8). More generally, we derive such equations for generic $-1 < p < q$. See Theorem 6.2. A characterization of these critical points is of great interest and if the (naturally $SL(n)$ invariant) class of critical points is rather small, this would lead to an alternative proof of the Gardner-Zhang [24] and Haddad-Ludwig [33] results, and potentially a resolution of the maximum of the ratio (1.8) for generic $-1 < p < q$, without restrictions on their values relative to n .

Finally, we remark here that our treatment of Theorem 1.1 is motivated by Lutwak-Xi-Yang-Zhang [51] and an earlier pioneering work [36] by Huang-Lutwak-Yang-Zhang. In particular, in Huang-Lutwak-Yang-Zhang [36], it was shown that, unlike quermassintegrals, the fundamental geometric invariants in the dual Brunn-Minkowski theory (started by Lutwak in the 1970s) allow for an Aleksandrov-type variational formula and that it leads to *dual curvature measures*, dual to Federer's curvature measures. The Minkowski problem for dual curvature measures, now known as the *dual Minkowski problem*, has been a focus in convex geometry and fully nonlinear elliptic PDEs for the last couple of years and has already led to a number of papers in a short period. See, for example, Böröczky-Henk-Pollehn [6], Chen-Chen-Li [15], Chen-Huang-Zhao [14], Chen-Li [17], Gardner-Hug-Weil-Xing-Ye [23], Henk-Pollehn [35], Li-Sheng-Wang [42], Liu-Lu [43], Zhao [70].

We refer the readers to the recent surveys Böröczky [4] and Huang-Yang-Zhang [37] for a more comprehensive account of works involving Minkowski-type problems.

2. PRELIMINARIES

First, we agree on some standard notations used in the current work. Throughout the paper, we will be working in the Euclidean space $\mathbb{R}^n$ with $n \geq 2$, where the standard inner product is denoted by $x \cdot y$ for $x, y \in \mathbb{R}^n$ and the Euclidean norm by $|x|$. The volume of the centered unit ball B_n in $\mathbb{R}^n$ will be written as ω_n , which implies the surface area of the centered unit sphere S^{n-1} is $n\omega_n$. We will use $\mathcal{H}^k$ for the k -dimensional Hausdorff measure. The space $C(S^{n-1})$ containing all continuous functions on S^{n-1} is equipped with the standard max norm; i.e., $\|f\|_{\max} = \max\{|f(u)| : u \in S^{n-1}\}$. We will use the superscript $+$ for positive functions, and the subscript e for even functions; that is, for example, the subspace $C_e^+(S^{n-1}) \subset C(S^{n-1})$ contains all positive even continuous functions on S^{n-1} . For each $u \in S^{n-1}$, we will write $u^\perp$ for the $(n-1)$ dimensional subspace orthogonal to u .

A subset K of $\mathbb{R}^n$ is called a *convex body* if it is a compact convex set with a non-empty interior. The collection of all convex bodies in $\mathbb{R}^n$ will be written as $\mathcal{K}^n$. The subcollection of only those containing the origin in the interior and of only those that are origin-symmetric will be denoted by $\mathcal{K}_o^n$ and $\mathcal{K}_e^n$, respectively. For simplicity, we will write $V(K)$ instead of $\mathcal{H}^n(K)$.

For general references to the theory of convex bodies, books such as [21] and [63] are recommended.

For a convex body K , we say its boundary ∂K is *line-free* in a direction $u \in S^{n-1}$ if it does not contain line segments parallel to u . It is well-known that ∂K is line-free in u for almost all $u \in S^{n-1}$.

Let K be a compact convex subset in $\mathbb{R}^n$. The *support function* $h_K : S^{n-1} \rightarrow \mathbb{R}$ is defined by

$$h_K(v) = \max\{x \cdot v : x \in K\}, \text{ for each } v \in S^{n-1}. \quad (2.1)$$

A compact subset $Q \subset \mathbb{R}^n$ is a *star-shaped set* with respect to a point $z \in Q$, if the intersection of each ray emanating from z and Q is always a (possibly degenerate, *i.e.*, containing only z) line segment. The radial function $\rho_{Q,z} : S^{n-1} \rightarrow \mathbb{R}$ is defined by

$$\rho_{Q,z}(u) = \max\{t \geq 0 : z + tu \in Q\}.$$

The subscript z will often be omitted if $z = o$. We say Q is a *star body* if $\rho_{Q,z}$ is positive and continuous. It is simple to see that a convex body K is a star-shaped set with respect to each $z \in K$ and a star body with respect to each $z \in \text{int } K$ (interior of K). We shall write

$$\Lambda_{K,z}^+ = \{u \in S^{n-1} : \rho_{K,z}(u) > 0\}.$$

Note that when $z \in \text{int } K$, we have $\Lambda_{K,z}^+ = S^{n-1}$ and since K is convex, for almost all $z \in \partial K$, the set $\Lambda_{K,z}^+$ is either an open or a closed hemisphere.

To make it simple in various settings, when we say star body (or star-shaped set) without mentioning the reference point z , it means $z = o$. We will also extend h_K to $\mathbb{R}^n$ homogeneously of degree 1 and ρ_Q to $\mathbb{R}^n \setminus \{o\}$ homogeneously of degree -1 .

Another important tool that we use in the current work to measure the length of a convex body in a certain direction is the *X-ray function* $X_K(\cdot, \cdot) : \mathbb{R}^n \times S^{n-1} \rightarrow \mathbb{R}$ given by

$$X_K(x, u) = \mathcal{H}^1(\{t \in \mathbb{R} : x + tu \in K\}).$$

At each boundary point z of $K \in \mathcal{K}^n$, a unit vector v is said to be an *outer unit normal* of K if $z \cdot v = h_K(v)$. If at z , there exists a unique outer unit normal, we will denote this by $\nu_K(z)$. By convexity, the Gauss map $\nu_K(z)$ is defined almost everywhere on ∂K . For each $\omega \subset S^{n-1}$, we will write $\nu_K^{-1}(\omega)$ for the preimage of ω under ν_K . The surface area measure of K , denoted by $S_{n-1}(K, \cdot)$, is a Borel measure on S^{n-1} given by

$$S_{n-1}(K, \omega) = \mathcal{H}^{n-1}(\nu_K^{-1}(\omega)), \quad \text{for each Borel subset } \omega \subset S^{n-1}. \quad (2.2)$$

Let K_i be a sequence of convex bodies in $\mathbb{R}^n$. We say K_i converges to a compact convex set $K \subset \mathbb{R}^n$ in the *Hausdorff metric* if $\|h_{K_i} - h_K\|_{\max} \rightarrow 0$. If K and K_i contain the origin in their interiors, then that K_i converges to K in the Hausdorff metric implies that

$$\|\rho_{K_i} - \rho_K\|_{\max} \rightarrow 0.$$

Let $\Omega \subset S^{n-1}$ be a compact subset not contained in any closed hemisphere of S^{n-1} . For each $f \in C(\Omega)$, define $[f]$ by

$$[f] = \bigcap_{v \in \Omega} \{x \in \mathbb{R}^n : v \cdot x \leq f(v)\}.$$

The compact convex set $[f]$ is called the *Wulff shape* or the *Aleksandrov body* generated by f . It is simple to see

$$h_{[f]} \leq f \quad \text{on } \Omega,$$

and if $K \in \mathcal{K}^n$, then $h_K \in C(S^{n-1})$ and

$$[h_K] = K.$$

The fundamental geometric functionals in the dual Brunn-Minkowski theory are dual quermassintegrals, and more generally dual mixed volumes. Let $K \in \mathcal{K}^n$ and L be a star body with respect to the origin. For $z \in K$ and $q > -1$, we may define the q -th dual mixed volume $\tilde{V}_q(K, L, z)$ as

$$\tilde{V}_q(K, L, z) = \frac{1}{n} \int_{\Lambda_{K,z}^+} \rho_{K,z}(u)^q \rho_L(u)^{n-q} du, \quad (2.3)$$

if the integral is finite. Note that when $z \in \text{int } K$, the integral in (2.3) is over the sphere S^{n-1} and it is obvious that the integral in (2.3) is finite as both $\rho_{K,z}$ and ρ_L are positive continuous functions on S^{n-1} . When $z \in \partial K$ and $q \geq 0$, the integral is finite since on $\Lambda_{K,z}^+$, the function $\rho_{K,z}$ is a bounded and positive function whereas ρ_L is a positive continuous function on S^{n-1} . The situation becomes much more delicate when $z \in \partial K$ and $q \in (-1, 0)$, as $\rho_{K,z}^q(u)$ becomes unbounded when $u \rightarrow \partial \Lambda_{K,z}^+$. We will prove in Corollary 3.4 that

$$\int_{\partial K} \tilde{V}_q(K, L, z) d\mathcal{H}^{n-1}(z) < \infty.$$

Consequently, the q -th dual mixed volume $\tilde{V}_q(K, L, z)$ is finite for almost all $z \in \partial K$.

When $L = B_n$, dual mixed volumes are often called *dual quermassintegrals* and denoted by

$$\tilde{V}_q(K, z) = \tilde{V}_q(K, L, z).$$

When $q > 0$, the dual quermassintegral can be written in the form of Riesz potential (by applying polar coordinates) or via the X -ray function (by appealing to the definitions of the X -ray function and the radial function):

$$\tilde{V}_q(K, z) = \frac{q}{n} \int_K |x - z|^{q-n} dx = \frac{1}{2n} \int_{S^{n-1}} X_K^q(z, u) du.$$

Let $K \in \mathcal{K}^n$ and $0 \neq q > -1$. The *radial q -th mean body* $R_q K$ of K is a star body (with respect to the origin) such that

$$\rho_{R_q K}(u) = \left(\frac{1}{V(K)} \int_K \rho_K(x, u)^q dx \right)^{\frac{1}{q}}. \quad (2.4)$$

In particular, when $q = n$, it can be seen that $V(R_n K) = V(K)$. When $q = \infty$, the body $R_\infty K$ is best known as the difference body $DK = K + (-K)$. It is important to note that although we choose not to mention the definition of $R_0 K$, it can be defined by taking the limit $q \rightarrow 0$. It is known (see [24]) that $R_q K$ is in fact convex for $q > 0$. Whether this remains true for all $q > -1$ is still open to this day.

A simple application of Jensen's inequality implies that for $-1 < p \leq q$, we have

$$R_p K \subset R_q K \subset R_\infty K = DK.$$

To study the asymptotic shape of $R_q K$ as $q \rightarrow -1^+$, a different normalization of $R_q K$ has to be considered. For $0 \neq q \geq -1$, the body $S_q K$ is a star body (with respect to the origin) such that

$$\rho_{S_q K}(u) = \left(\int_{u^\perp} X_K(y, u)^{q+1} d\mathcal{H}^{n-1}(y) \right)^{\frac{1}{q}}.$$

It is simple to see that $S_\infty K = DK$ and $S_{-1} K = \Pi^* K$ where $\Pi^* K$ is the *polar projection body* whose radial function is given by

$$\rho_{\Pi^* K}^{-1}(u) = \mathcal{H}^{n-1}(K|u^\perp), \quad (2.5)$$

where $K|u^\perp$ is the image of the orthogonal projection of K onto $u^\perp$. By [24, Theorem 2.2], the radial q th mean body $R_q K$ and $S_q K$ when $q > -1$ are related to each other by a proper renormalization:

$$R_q K = [(q+1)V(K)]^{-\frac{1}{q}} S_q K. \quad (2.6)$$

We will also need the fact that when $q \rightarrow -1^+$, the radial function $\rho_{S_q K}$ converges to $\rho_{\Pi^* K}$ uniformly on S^{n-1} . This shows that, up to a proper rescaling, the shape of $R_q K$ converges to that of $\Pi^* K$ when $q \rightarrow -1^+$.

It is crucial to note that the constructions above are natural objects in affine geometry: for $\phi \in \text{SL}(n)$ and $x_0 \in \mathbb{R}^n$, we have

$$R_q(\phi K + x_0) = \phi R_q K, \quad S_q(\phi K + x_0) = \phi S_q K, \quad \text{and} \quad \Pi^*(\phi K + x_0) = \phi \Pi^* K.$$

They are also homogeneous: for $\lambda > 0$, we have

$$R_q(\lambda K) = \lambda R_q K, \quad S_q(\lambda K) = \lambda^{\frac{n+q}{q}} S_q K, \quad \text{and} \quad \Pi^*(\lambda K) = \lambda^{1-n} \Pi^* K.$$

For easy reference, we repeat the following definition.

Definition 2.1. *Let K be a convex body in $\mathbb{R}^n$ and $0 \neq q \geq -1$. The q -th fractional integral affine surface area is given by*

$$\Phi_q(K) = V(S_q K)^{\frac{q}{n}} = \left(\frac{1}{n} \int_{S^{n-1}} \left(\int_{u^\perp} X_K(y, u)^{q+1} d\mathcal{H}^{n-1}(y) \right)^{\frac{n}{q}} du \right)^{\frac{q}{n}}.$$

By [24, Lemma 2.1], an alternative and useful way of writing $\Phi_q(K)$ is

$$\Phi_q(K) = \left(\frac{1}{n} \int_{S^{n-1}} \left((q+1) \int_K \rho_K(x, u)^q dx \right)^{\frac{n}{q}} du \right)^{\frac{q}{n}}.$$

When $q = n$, we have

$$\Phi_n(K) = \frac{n+1}{n} \int_{S^{n-1}} \int_K \rho_K(x, u)^n dx du = (n+1) \int_K \frac{1}{n} \int_{S^{n-1}} \rho_K(x, u)^n du dx = (n+1)V(K)^2.$$

At the other end of the spectrum, when $q = -1$, we have

$$\Phi_{-1}(K) = \Phi(K).$$

It is simple to see that $\Phi_q(K)$ is homogeneous in K of degree $q+n$, and it is translation-invariant and $\text{SL}(n)$ -invariant in K . We will also need to use the fact that $\Phi_q(\cdot)$ is a continuous functional on $\mathcal{K}^n$ with respect to the Hausdorff metric.

3. FRACTIONAL INTEGRAL AFFINE SURFACE AREA AND MEASURE

This section is devoted to showing properties of the following family of measures and that they in fact arise from properly differentiating fractional integral affine surface area.

Definition 3.1. *Let $K \in \mathcal{K}^n$ and $0 \neq q > -1$. The q -th fractional affine area measure is a Borel measure on S^{n-1} given by*

$$F_q^{(a)}(K, \omega) = (q+1) \int_{\nu_K^{-1}(\omega)} \tilde{V}_q(K, S_q K, z) d\mathcal{H}^{n-1}(z).$$

for every Borel subset $\omega \subset S^{n-1}$.

Using the definitions of X -ray function, radial function, and dual mixed volume, we may rewrite the definition of $F_q^{(a)}$ as

$$F_q^{(a)}(K, \omega) = \frac{q+1}{2n} \int_{\nu_K^{-1}(\omega)} \left(\int_{S^{n-1}} \rho_{S_q K}^{n-q}(u) X_K(z, u)^q du \right) d\mathcal{H}^{n-1}(z).$$

Note that it is clear from its definition that $F_q^{(a)}(K, \cdot)$ is absolutely continuous with respect to the surface area measure $S_{n-1}(K, \cdot)$ of K . In particular, if K is strictly convex and has everywhere positive Gauss curvature, then

$$\frac{dF_q^{(a)}(K, v)}{dv} = \frac{q+1}{2n} \left(\int_{S^{n-1}} \rho_{S_q K}^{n-q}(u) X_K(\nabla_{S^{n-1}} h_K(v) + h_K(v)v, u)^q du \right) \det(\nabla_{S^{n-1}}^2 h_K(v) + h_K(v)I).$$

It is important to note that even with sufficient regularity, the fractional affine area measure is a *nonlocal* notion; that is, just from the shape of K around a neighborhood of $\nu_K^{-1}(v)$, it is not possible to determine $F_q^{(a)}(K, \cdot)$ in any neighborhood of v .

It follows from the definition that $F_q^{(a)}(K, \cdot)$ is homogeneous in K of degree $\frac{n(q+n)}{q} - 1$.

One of the major goals in this section is to prove Theorem 1.1, for which Lemma 3.2, Theorem 3.3, and Theorem 3.7 are essential.

Lemma 3.2. (Integration by parts for concave functions.) *Let $q > -1$ and $K \in \mathcal{K}^n$. If $F : K \rightarrow [0, \infty)$ is concave, continuous, and $F > 0$ in $\text{int } K$ and vanishes on ∂K , then*

$$\int_K x \cdot \nabla (F^{q+1}(x)) dx = -n \int_K F(x)^{q+1} dx.$$

Proof. We first prove the 1-dimensional case and then the general cases.

- (1) *The 1-dimensional case ($n = 1$).* Without loss of generality, suppose $K = [a, b]$ for $0 < a < b$. It is clear that $xF(x)^{q+1}$ is Lipschitz and hence absolutely continuous on the closed interval $[a + \varepsilon, b - \varepsilon]$ where $\varepsilon > 0$ is arbitrarily chosen. Then, we have

$$\begin{aligned} & (q+1) \int_{a+\varepsilon}^{b-\varepsilon} xF(x)^q F'(x) dx \\ &= (b-\varepsilon)F(b-\varepsilon)^{q+1} - (a+\varepsilon)F(a+\varepsilon)^{q+1} - \int_{a+\varepsilon}^{b-\varepsilon} F(x)^{q+1} dx. \end{aligned} \tag{3.1}$$

Note that since F is a nonnegative concave function on $[a, b]$ and vanishes at $x = a, b$, it must first increase and then decrease. Therefore, by the monotone convergence theorem,

the fact that F is continuous with $F(a) = F(b) = 0$, we may take the limit as $\varepsilon \rightarrow 0^+$ in (3.1) and get

$$\int_a^b x \cdot [F^{q+1}(x)]' dx = (q+1) \int_a^b x F(x)^q F'(x) dx = - \int_a^b F(x)^{q+1} dx.$$

- (2) *The n -dimensional case.* Let $e_1, \dots, e_n$ be the canonical basis of $\mathbb{R}^n$, $(x_1, \dots, x_n)$ be the coordinates of x , and $\partial_j F(x)$ be the partial differential of F with respect to x_j . It suffices to show

$$(q+1) \int_K x_1 F(x)^q \partial_1 F(x) dx = - \int_K F(x)^{q+1} dx. \quad (3.2)$$

Write $x = (x_1, x')$. Since F is nonnegative, we have

$$\int_K F(x)^{q+1} dx = \int_{K|e_1^\perp} \left(\int_{(x_1, x') \in K} F(x_1, x')^{q+1} dx_1 \right) dx'.$$

By the 1-dimensional case, for each $x' \in K|e_1^\perp$, we have

$$(q+1) \int_{(x_1, x') \in K} x_1 F(x_1, x')^q \partial_1 F(x_1, x') dx_1 = - \int_{(x_1, x') \in K} F(x_1, x')^{q+1} dx_1.$$

Integrating with respect to $x' \in K|e_1^\perp$, we conclude (3.2). □

For notational simplicity, we will write for $q > 0$ and $u \in S^{n-1}$,

$$I_q(K, u) = \int_{u^\perp} X_K(y, u)^q d\mathcal{H}^{n-1}(y).$$

Since $X_{\lambda K}(y, u) = \lambda X_K(y/\lambda, u)$ for each $\lambda > 0$, we have

$$I_{q+1}(\lambda K, u) = \int_{u^\perp} X_{\lambda K}(y, u)^{q+1} d\mathcal{H}^{n-1}(y) = \lambda^{n+q} I_{q+1}(K, u). \quad (3.3)$$

Recall that

$$\rho_{S_q K}(u) = I_{q+1}(K, u)^{\frac{1}{q}}.$$

The next theorem contains a divergence-type integral formula that is critical in showing Theorem 1.1.

Theorem 3.3. *Let $K \in \mathcal{K}^n$, and $q > -1$. If ∂K is line-free in $u \in S^{n-1}$, then*

$$I_{q+1}(K, u) = \frac{q+1}{n+q} \int_{\partial K} X_K(z, u)^q \left(z \cdot \nu_K(z) \right) d\mathcal{H}^{n-1}(z). \quad (3.4)$$

Proof. There exist $f_K, g_K : K|u^\perp \rightarrow \mathbb{R}$ such that

$$K = \left\{ x + tu : f_K(x) \leq t \leq g_K(x), \quad x \in K|u^\perp \right\}.$$

Since K is convex, both f_K and g_K are locally Lipschitz. Using the fact that ∂K is line-free in u , for every integrable function $F : \partial K \rightarrow \mathbb{R}$,

$$\begin{aligned} \int_{\partial K} F(z) d\mathcal{H}^{n-1}(z) &= \int_{K|u^\perp} F\left(x + g_K(x)u\right) \sqrt{1 + |\nabla g_K(x)|^2} d\mathcal{H}^{n-1}(x) \\ &\quad + \int_{K|u^\perp} F\left(x + f_K(x)u\right) \sqrt{1 + |\nabla f_K(x)|^2} d\mathcal{H}^{n-1}(x). \end{aligned} \quad (3.5)$$

For almost all $x \in K|u^\perp$, at the boundary points $x + g_K(x)u$ and $x + f_K(x)u$, we have

$$\nu_K(x + g_K(x)u) = \frac{-\nabla g_K(x) + u}{\sqrt{1 + |\nabla g_K(x)|^2}}, \quad \text{and} \quad \nu_K(x + f_K(x)u) = \frac{\nabla f_K(x) - u}{\sqrt{1 + |\nabla f_K(x)|^2}}.$$

Substituting this and $F(z) = X_K(z, u)^q(z \cdot \nu_K(z))$ into (3.5), and using that $X_K(x, u) = g_K(x) - f_K(x)$ for $x \in K|u^\perp$, we have

$$\begin{aligned} & \int_{\partial K} X_K(z, u)^q (z \cdot \nu_K(z)) d\mathcal{H}^{n-1}(z) \\ &= \int_{K|u^\perp} X_K(x, u)^q (x \cdot \nabla f_K(x) - x \cdot \nabla g_K(x) + g_K(x) - f_K(x)) d\mathcal{H}^{n-1}(x) \\ &= \int_{K|u^\perp} \left[X_K(x, u)^{q+1} - \frac{1}{q+1} x \cdot \nabla (X_K(x, u)^{q+1}) \right] d\mathcal{H}^{n-1}(x). \end{aligned} \quad (3.6)$$

Since ∂K is line-free in u , $X_K(x, u)$ vanishes on the relative boundary of $K|u^\perp$ and is positive in the relative interior of $K|u^\perp$. It follows from Lemma 3.2 that

$$\int_{K|u^\perp} \frac{1}{q+1} x \cdot \nabla (X_K(x, u)^{q+1}) d\mathcal{H}^{n-1}(x) = \frac{1-n}{q+1} \int_{K|u^\perp} X_K(x, u)^{q+1} d\mathcal{H}^{n-1}(x).$$

This together with (3.6) gives (3.4). $\square$

While it is clear that $F_q^{(a)}(K, \cdot)$ defines a finite measure when $q > 0$, it is less clear why it is the case when $q \in (-1, 0)$. We now take a quick detour to show this fact. Note that this fact can be derived from [51, Lemma 4.9]. However, the treatment there involves introducing a parameter ε to avoid singularities in the integral and then taking limit as $\varepsilon \rightarrow 0$. Our treatment here is much more direct and it follows immediately from Theorem 3.3.

Corollary 3.4. *Let $K \in \mathcal{K}^n$, L be a star body, and $q > -1$. Then,*

$$\int_{\partial K} \tilde{V}_q(K, L, z) d\mathcal{H}^{n-1}(z) < \infty.$$

Proof. Since L is a star body, there exists $C_L > 0$ such that $1/C_L < \rho_L < C_L$ on S^{n-1} . By the definition of dual mixed volume, it is sufficient to show

$$\int_{\partial K} \tilde{V}_q(K, z) d\mathcal{H}^{n-1}(z) < \infty.$$

Since the integral above is invariant under the map $K \mapsto K + x_0$ for every $x_0 \in \mathbb{R}^n$, we may assume $K \in \mathcal{K}_o^n$ and consequently, there exists $c_0 > 0$ such that

$$z \cdot \nu_K(z) > c_0, \quad (3.7)$$

for almost all $z \in \partial K$.

By the definitions of dual mixed volume, the X -ray function, (3.7), the Fubini's theorem, Theorem 3.3, we have

$$\begin{aligned}
\int_{\partial K} \tilde{V}_q(K, z) d\mathcal{H}^{n-1}(z) &= \frac{1}{n} \int_{\partial K} \int_{\Lambda_{K,z}^+} \rho_{K,z}^q(u) du d\mathcal{H}^{n-1}(z) \\
&= \frac{1}{2n} \int_{\partial K} \int_{S^{n-1}} X_K^q(z, u) du d\mathcal{H}^{n-1}(z) \\
&\leq \frac{1}{2nc_0} \int_{\partial K} \int_{S^{n-1}} X_K^q(z, u) (z \cdot \nu_K(z)) du d\mathcal{H}^{n-1}(z) \\
&= \frac{1}{2nc_0} \int_{S^{n-1}} \int_{\partial K} X_K^q(z, u) (z \cdot \nu_K(z)) d\mathcal{H}^{n-1}(z) du \\
&= \frac{n+q}{2nc_0(q+1)} \int_{S^{n-1}} I_{q+1}(K, u) du,
\end{aligned}$$

where the integral on the last line is finite since K is compact. (Note that $(q+1)$ is always positive.) $\square$

Since $S_q K$ is a star body when $K \in \mathcal{K}^n$, Corollary 3.4, in particular, implies that $F_q^{(a)}(K, \cdot)$ is a finite measure. We record this fact in the next corollary.

Corollary 3.5. *Let $K \in \mathcal{K}^n$ and $0 \neq q > -1$. The q -th fractional affine area measure of K , $F_q^{(a)}(K, \cdot)$, is a finite measure on S^{n-1} .*

The following lemma providing the derivative of the X -ray function was proved in [51].

Lemma 3.6 ([51]). *Let K_t be as defined in Theorem 1.1. If $u \in S^{n-1}$, then for almost all x in the interior of $K_0|u^\perp$,*

$$\left. \frac{dX_{K_t}(x, u)}{dt} \right|_{t=0} = \frac{g(\nu_{K_0}(z^+))}{u \cdot \nu_{K_0}(z^+)} - \frac{g(\nu_{K_0}(z^-))}{u \cdot \nu_{K_0}(z^-)},$$

where z^+ and z^- are the upper and lower points of $\partial K_0 \cap (x + \mathbb{R}u)$.

We need the following generalized dominated convergence theorem. Suppose f_k, ϕ_k, f, ϕ are integrable functions in a measure space with $f_k \rightarrow f$, $\phi_k \rightarrow \phi$, and $|f_k| \leq \phi_k$ pointwise almost everywhere. If $\int \phi_k \rightarrow \int \phi$, then $\int f_k \rightarrow \int f$.

Theorem 3.7. *Let $q > -1$ and K_t be as defined in Theorem 1.1. Then, for almost all $u \in S^{n-1}$,*

$$\left. \frac{d}{dt} \right|_{t=0} I_{q+1}(K_t, u) = (q+1) \int_{\partial K_0} g(\nu_{K_0}(z)) X_{K_0}(z, u)^q d\mathcal{H}^{n-1}(z).$$

Proof. Without loss of generality, assume K_0 contains the origin in its interior. Let $u \in S^{n-1}$ be a direction such that ∂K_0 is line-free in u . Recall that such u 's constitute almost all points in S^{n-1} .

Since g is continuous on S^{n-1} , there is a constant $c > 0$, such that

$$(1 - c|t|)K_0 \subset K_t \subset (1 + c|t|)K_0. \tag{3.8}$$

Denote

$$\phi_t(x) = \frac{X_{(1+c|t|)K_0}(x, u)^{q+1} - X_{(1-c|t|)K_0}(x, u)^{q+1}}{|t|}.$$

By (3.3),

$$\int_{u^\perp} \phi_t(x) d\mathcal{H}^{n-1}(x) = \frac{(1 + c|t|)^{n+q} - (1 - c|t|)^{n+q}}{|t|} I_{q+1}(K_0, u).$$

This, together with Theorem 3.3, gives

$$\lim_{t \rightarrow 0} \int_{u^\perp} \phi_t(x) dx = 2c(q+1) \int_{\partial K_0} X_{K_0}(z, u)^q \left(z \cdot \nu_{K_0}(z) \right) d\mathcal{H}^{n-1}(z). \quad (3.9)$$

By Lemma 3.6, for almost all x in the interior of $K_0|u^\perp$, we have

$$\lim_{t \rightarrow 0} \phi_t(x) = 2c(q+1) X_{K_0}(x, u)^q \left(\frac{h_{K_0}(\nu_{K_0}(z^+))}{u \cdot \nu_{K_0}(z^+)} - \frac{h_{K_0}(\nu_{K_0}(z^-))}{u \cdot \nu_{K_0}(z^-)} \right).$$

It is simple to see from the choice of $\phi_t(x)$ that if $x \in u^\perp$ is not in $K_0|u^\perp$, then $\lim_{t \rightarrow 0} \phi_t(x) = 0$. It follows then from a change-of-variable and (3.9) that

$$\int_{u^\perp} \lim_{t \rightarrow 0} \phi_t(x) dx = 2c(q+1) \int_{\partial K_0} X_{K_0}(z, u)^q \left(z \cdot \nu_{K_0}(z) \right) d\mathcal{H}^{n-1}(z) = \lim_{t \rightarrow 0} \int_{u^\perp} \phi_t(x) dx.$$

On the other hand, by (3.8), we have

$$\left| \frac{X_{K_t}(x, u)^{q+1} - X_{K_0}(x, u)^{q+1}}{t} \right| \leq \phi_t(x).$$

Now, the conclusion follows from the generalized dominated convergence theorem— ϕ_t being the sequence of dominating functions—and the following equation guaranteed by Lemma 3.6

$$\lim_{t \rightarrow 0} \frac{X_{K_t}(x, u)^{q+1} - X_{K_0}(x, u)^{q+1}}{t} = (q+1) X_{K_0}(x, u)^q \left(\frac{g(\nu_{K_0}(z^+))}{u \cdot \nu_{K_0}(z^+)} - \frac{g(\nu_{K_0}(z^-))}{u \cdot \nu_{K_0}(z^-)} \right),$$

for almost all x in the interior of $K_0|u^\perp$. $\square$

We are now ready to provide a proof of Theorem 1.1.

Proof of Theorem 1.1. Since everything is translation invariant, we may, by adding or subtracting a linear function from h_0 (or, equivalently, translate K_0), assume that $h_0 \in C^+(S^{n-1})$ (or, equivalently, K_0 contains the origin as an interior point).

By Theorem 3.7, $I_{q+1}(K_t, u)$ is differentiable for almost all $u \in S^{n-1}$. Now, we want to prove

$$\frac{d}{dt} \Big|_{t=0} \left(\int_{S^{n-1}} I_{q+1}(K_t, u)^{\frac{n}{q}} du \right) = \int_{S^{n-1}} \left(\frac{d}{dt} \Big|_{t=0} I_{q+1}(K_t, u)^{\frac{n}{q}} \right) du. \quad (3.10)$$

By (3.8) and (3.3), we know that

$$\begin{aligned} \left| \frac{I_{q+1}(K_t, u)^{\frac{n}{q}} - I_{q+1}(K_0, u)^{\frac{n}{q}}}{t} \right| &\leq \left| \frac{I_{q+1}((1 + c|t|)K_0, u)^{\frac{n}{q}} - I_{q+1}((1 - c|t|)K_0, u)^{\frac{n}{q}}}{t} \right| \\ &= \left| \frac{(1 + c|t|)^{\frac{n(n+q)}{q}} - (1 - c|t|)^{\frac{n(n+q)}{q}}}{t} \right| I_{q+1}(K_0, u)^{\frac{n}{q}}. \end{aligned}$$

Note that there exists $C_0 > 0$ such that for all sufficiently small $|t| \neq 0$,

$$\left| \frac{(1 + c|t|)^{\frac{n(n+q)}{q}} - (1 - c|t|)^{\frac{n(n+q)}{q}}}{t} \right| \leq C_0.$$

Note also that

$$\int_{S^{n-1}} I_{q+1}(K_0, u)^{\frac{n}{q}} du = n\Phi_q(K_0)^{\frac{n}{q}} < \infty.$$

Hence, (3.10) follows from the dominated convergence theorem. Consequently, by Theorem 3.7, we have

$$\begin{aligned} & \left. \frac{d}{dt} \right|_{t=0} \left(\int_{S^{n-1}} I_{q+1}(K_t, u)^{\frac{n}{q}} du \right) \\ &= \frac{n(q+1)}{q} \int_{S^{n-1}} I_{q+1}(K_0, u)^{\frac{n-q}{q}} \int_{\partial K_0} g(\nu_{K_0}(z)) X_{K_0}(z, u)^q d\mathcal{H}^{n-1}(z) du \\ &= \frac{n(q+1)}{q} \int_{\partial K_0} g(\nu_{K_0}(z)) \int_{S^{n-1}} I_{q+1}(K_0, u)^{\frac{n-q}{q}} X_{K_0}(z, u)^q du d\mathcal{H}^{n-1}(z) \\ &= \frac{n(q+1)}{q} \int_{\partial K_0} g(\nu_{K_0}(z)) \int_{S^{n-1}} \rho_{S_q K_0}(u)^{n-q} X_{K_0}(z, u)^q du d\mathcal{H}^{n-1}(z) \\ &= \frac{2n^2}{q} \int_{\Omega} g(v) dF_q^{(a)}(K_0, v), \end{aligned}$$

where the last line follows from the definition of $F_q^{(a)}(K_0, \cdot)$ and the fact that $\nu_{K_0}(z) \in \Omega$ for almost all $z \in \partial K_0$. Now, Theorem 1.1 follows immediately from the chain rule. $\square$

Remark 3.8. *In the above proof, one can replace the power $\frac{n}{q}$ by 1 without incurring much change. This will recover [51, Theorem 5.5].*

Motivated by the success of the L_p Brunn-Minkowski theory initiated by Lutwak [49, 50], we now introduce the L_p version of the fractional affine area measure.

Definition 3.9. *Let $K \in \mathcal{K}_o^n$, $0 \neq q > -1$, and $p \in \mathbb{R}$. The L_p fractional affine area measure of q -th order is given by*

$$F_{p,q}^{(a)}(K, \cdot) = h_K^{1-p} dF_q^{(a)}(K, \cdot).$$

The following theorem follows directly from Theorem 1.1.

Theorem 3.10. *Let $0 \neq q > -1$, $p \in \mathbb{R}$, $\Omega \subset S^{n-1}$ be a compact subset not contained in any closed hemisphere, and $h_0 \in C^+(\Omega)$, $g \in C(\Omega)$. Suppose that for sufficiently small $\delta > 0$ and $|t| < \delta$, the Wulff shape $K_t = [h_t]$ where on Ω*

$$\begin{aligned} h_t^p &= h_0^p + tg + o(t, \cdot), & \text{if } p \neq 0 \\ \log h_t &= \log h_0 + tg + o(t, \cdot), & \text{if } p = 0 \end{aligned}$$

is a convex body in $\mathbb{R}^n$ containing the origin as an interior point. Here $o(t, \cdot)/t \rightarrow 0$ uniformly on Ω as $t \rightarrow 0$. Then,

$$\begin{aligned} \left. \frac{d}{dt} \right|_{t=0} \Phi_q(K_t) &= \frac{2}{p} \Phi_q^{\frac{q-n}{q}}(K_0) \int_{\Omega} g(v) dF_{p,q}^{(a)}(K_0, v), \quad \text{if } p \neq 0, \\ \left. \frac{d}{dt} \right|_{t=0} \Phi_q(K_t) &= 2 \Phi_q^{\frac{q-n}{q}}(K_0) \int_{\Omega} g(v) dF_{0,q}^{(a)}(K_0, v), \quad \text{if } p = 0. \end{aligned}$$

By homogeneity of Φ_q , we derive the following integral formula.

Theorem 3.11. *Let $K \in \mathcal{K}^n$ and $0 \neq q > -1$. Then*

$$\int_{S^{n-1}} h_K(v) dF_q^{(a)}(K, v) = \frac{q+n}{2} \Phi_q^{\frac{n}{q}}(K).$$

Proof. Let $\Omega = S^{n-1}$ and $h_0 = g = h_K$ in Theorem 1.1. We conclude that

$$\left. \frac{d}{dt} \right|_{t=0} \Phi_q((1+t)K) = 2 \Phi_q^{\frac{q-n}{q}}(K) \int_{S^{n-1}} h_K(v) dF_q^{(a)}(K, v).$$

On the other hand,

$$\left. \frac{d}{dt} \right|_{t=0} \Phi_q((1+t)K) = \left. \frac{d}{dt} \right|_{t=0} (1+t)^{q+n} \Phi_q(K) = (q+n) \Phi_q(K).$$

By combining the above two equations, we have the desired integral formula. $\square$

A direct consequence of Theorem 3.11 is

Corollary 3.12. *Let $K \in \mathcal{K}_o^n$ and $0 \neq q > -1$. Then*

$$F_{0,q}^{(a)}(K, S^{n-1}) = \frac{q+n}{2} \Phi_q^{\frac{n}{q}}(K).$$

By translation-invariance of Φ_q , we can immediately conclude that $F_q^{(a)}(K, \cdot)$ is translation-invariant in K .

Proposition 3.13. *Let $K \in \mathcal{K}^n$ and $0 \neq q > -1$. Then*

$$F_q^{(a)}(K, \cdot) = F_q^{(a)}(K + x_0, \cdot),$$

for every $x_0 \in \mathbb{R}^n$.

Proof. Let $\Omega = S^{n-1}$ and $h_0 = h_K$ in Theorem 1.1. We have

$$\left. \frac{d}{dt} \right|_{t=0} \Phi_q(K_t) = 2 \Phi_q^{\frac{q-n}{q}}(K) \int_{S^{n-1}} g(v) dF_q^{(a)}(K, \cdot).$$

Applying the same formula, this time with $h_0 = h_{K+x_0}$, we have

$$\left. \frac{d}{dt} \right|_{t=0} \Phi_q(K_t + x_0) = 2 \Phi_q^{\frac{q-n}{q}}(K + x_0) \int_{S^{n-1}} g(v) dF_q^{(a)}(K + x_0, \cdot).$$

The desired result now follows from the fact that $\Phi_q(K_t) = \Phi_q(K_t + x_0)$ and the fact that the above two formulas hold for all $g \in C(S^{n-1})$. $\square$

The next proposition includes the property $F_q^{(a)}(K, \cdot)$ inherited from the affine invariance of $\Phi_q(K)$.

Lemma 3.14. *Let $K \in \mathcal{K}^n$ and $0 \neq q > -1$. If $\phi \in \text{SL}(n)$, then for every $f \in C(S^{n-1})$,*

$$\int_{S^{n-1}} f(v) dF_q^{(a)}(K, v) = \int_{S^{n-1}} |\phi^t v| f\left(\frac{\phi^t v}{|\phi^t v|}\right) dF_q^{(a)}(\phi K, v).$$

Proof. Let $L \in \mathcal{K}^n$. By Theorem 1.1, we have

$$\left. \frac{d}{dt} \right|_{t=0^+} \Phi_q(K + tL) = 2 \Phi_q^{\frac{q-n}{q}}(K) \int_{S^{n-1}} h_L(v) dF_q^{(a)}(K, v).$$

On the other hand,

$$\begin{aligned} \left. \frac{d}{dt} \right|_{t=0^+} \Phi_q(K + tL) &= \left. \frac{d}{dt} \right|_{t=0^+} \Phi_q(\phi K + t\phi L) \\ &= 2 \Phi_q^{\frac{q-n}{q}}(\phi K) \int_{S^{n-1}} h_{\phi L}(v) dF_q^{(a)}(\phi K, v) \\ &= 2 \Phi_q^{\frac{q-n}{q}}(K) \int_{S^{n-1}} h_L(\phi^t v) dF_q^{(a)}(\phi K, v). \end{aligned}$$

Combining the above inequalities, we have

$$\int_{S^{n-1}} h_L(v) dF_q^{(a)}(K, v) = \int_{S^{n-1}} h_L(\phi^t v) dF_q^{(a)}(\phi K, v) = \int_{S^{n-1}} |\phi^t v| h_L\left(\frac{\phi^t v}{|\phi^t v|}\right) dF_q^{(a)}(\phi K, v),$$

for all $L \in \mathcal{K}^n$. Since every function in $C^2(S^{n-1})$ can be represented as the difference of support functions of two convex bodies and that $C^2(S^{n-1})$ is dense in $C(S^{n-1})$ (see [63, Lemma 1.7.8]), using the fact that $F_q^{(a)}(K, \cdot)$ and $F_q^{(a)}(\phi K, \cdot)$ are finite measures (Corollary 3.5), we get the desired result. $\square$

Given a finite Borel measure μ on S^{n-1} and $\phi \in \text{SL}(n)$, the affine image $\phi\mu$, as defined in [9], is given by

$$\phi\mu(\omega) = \mu(\langle \phi^{-1}\omega \rangle), \quad \text{for each Borel subset } \omega \subset S^{n-1}, \quad (3.11)$$

where $\langle \phi^{-1}\omega \rangle = \{\phi^{-1}v/|\phi^{-1}v| : v \in \omega\}$. Let $\mu(K, \cdot)$ be a family of finite Borel measures on S^{n-1} parametrized by convex bodies K . We say $\mu(K, \cdot)$ is $\text{SL}(n)$ -contravariant in K , if $\mu(\phi K, \cdot) = \phi^{-t}\mu(K, \cdot)$ for each $\phi \in \text{SL}(n)$. Lemma 3.14 shows that $F_{0,q}^{(a)}(K, \cdot)$ is $\text{SL}(n)$ -contravariant.

Theorem 3.15. *Let $K \in \mathcal{K}_o^n$ and $0 \neq q > -1$. If $\phi \in \text{SL}(n)$, then*

$$F_{0,q}^{(a)}(\phi K, \cdot) = \phi^{-t} F_{0,q}^{(a)}(K, \cdot).$$

Proof. Let $f \in C(S^{n-1})$ be arbitrary. By Lemma 3.14, the homogeneity of support functions we have

$$\begin{aligned} \int_{S^{n-1}} f(v) dF_{0,q}^{(a)}(K, v) &= \int_{S^{n-1}} |\phi^t v| f\left(\frac{\phi^t v}{|\phi^t v|}\right) h_K\left(\frac{\phi^t v}{|\phi^t v|}\right) dF_q^{(a)}(\phi K, v) \\ &= \int_{S^{n-1}} f\left(\frac{\phi^t v}{|\phi^t v|}\right) h_K(\phi^t v) dF_q^{(a)}(\phi K, v) \\ &= \int_{S^{n-1}} f\left(\frac{\phi^t v}{|\phi^t v|}\right) dF_{0,q}^{(a)}(\phi K, v). \end{aligned}$$

Since f is arbitrary, this concludes the proof. $\square$

The following theorem shows that $F_q^{(a)}(K, \cdot)$ has its centroid at the origin.

Theorem 3.16. *Let $K \in \mathcal{K}^n$ and $0 \neq q > -1$. Then,*

$$\int_{S^{n-1}} v dF_q^{(a)}(K, v) = o.$$

Proof. Let $x_0 \in \mathbb{R}^n$ be arbitrary. By Theorem 3.11 and the translation-invariance of Φ_q , we have

$$\int_{S^{n-1}} h_K(v) dF_q^{(a)}(K, v) = \int_{S^{n-1}} h_{K+x_0}(v) dF_q^{(a)}(K+x_0, v) = \int_{S^{n-1}} (h_K(v) + x_0 \cdot v) dF_q^{(a)}(K+x_0, v).$$

Using Proposition 3.13, we have

$$\int_{S^{n-1}} x_0 \cdot v dF_q^{(a)}(K, v) = 0.$$

Since x_0 is arbitrary, we arrive at the desired consequence. $\square$

4. THE DIFFERENTIAL OF THE INTEGRAL AFFINE SURFACE AREA

In this section, we will derive variational formulas for the classical integral affine surface area and define the *affine $(n-2)$ -area measure* of a convex body. We shall see below that with additional regularity assumptions, the fractional affine area measure $F_q^{(a)}(K, \cdot)$ converges to the affine area measure as $q \rightarrow -1^+$.

With sufficient regularity assumptions, we will write $\kappa_K(z)$ for the Gauss curvature of ∂K at the boundary point z and $\kappa_K(z, u)$ for the normal curvature for each unit vector u in the tangent space.

Lemma 4.1. *Suppose $K \in \mathcal{K}^n$ has C^2 boundary and everywhere positive Gauss curvature. Let $x \in \partial K$, v be the normal vector at x , and $u \in v^\perp \cap S^{n-1}$. Then,*

$$\kappa_{K|u^\perp}(x|u^\perp) = \frac{\kappa_K(x)}{\kappa_K(x, u)}.$$

Proof. Let $h : \mathbb{R}^n \rightarrow \mathbb{R}$ be the support function of K . Let $Dh(v) \in \mathbb{R}^n$ and $D^2h(v) : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the Euclidean gradient and Hessian of h , respectively. Since h is 1-homogeneous, $D^2h(v)$ is self-adjoint and v is an eigenvector with eigenvalue 0. Moreover, $Dh(v) = x$, and $D^2h(v)$, when restricted to $v^\perp$, is exactly the reverse Weingarten map. Then,

$$\det(D_{v^\perp}^2 h(v)) = \det(D^2h(v)|_{v^\perp}) = \text{reciprocal Gauss curvature of } \partial K \text{ at } x.$$

Here $D_{v^\perp}^2 h(v)$ denotes the restriction of D^2h to $v^\perp$, and $D_{v^\perp}^2 h(v)$ denotes the Euclidean Hessian of h taken in $v^\perp$. Based on the argument above, they are equivalent.

The same argument in $u^\perp$, in addition to the fact that the support function of $K|u^\perp$ is precisely the restriction of h to $u^\perp$, imply

$$\det(D_{u^\perp \cap v^\perp}^2 h(v)) = \text{reciprocal Gauss curvature of } \partial(K|u^\perp) \text{ at } x|u^\perp. \quad (4.1)$$

Here $D_{u^\perp \cap v^\perp}^2 h(v)$ is the Euclidean Hessian of h taken in $u^\perp \cap v^\perp$.

Recall that the positive eigenvalues of $D^2h(v)$ are the principal radii of curvature $1/\kappa_1, \dots, 1/\kappa_{n-1}$, and the associated eigenvectors $e_1, \dots, e_{n-1}$ are the principal directions. Denote $A = D^2h(v)|_{v^\perp}$, $B = D_{u^\perp \cap v^\perp}^2 h(v)$. Consider the matrix presentation of A and u , with respect to $e_1, \dots, e_{n-1}$. Obviously, A is exactly the diagonal matrix

$$A = \text{diag}(1/\kappa_1, \dots, 1/\kappa_{n-1}).$$

Denote $u = (u_1, \dots, u_{n-1})^t$ to be the representation of $u \in v^\perp$. Let $w_1, \dots, w_{n-2} \in \mathbb{R}^{n-1}$ be an orthonormal basis of $u^\perp \cap v^\perp$. Since B is just the Euclidean Hessian of h in $u^\perp \cap v^\perp$, we see that

$\det(B)$ is exactly the determinant of $(w_i^t A w_j)_{i,j}$. Denote $Q = (w_1, \dots, w_{n-2})$ be the $(n-1) \times (n-2)$ matrix generated by $w_1, \dots, w_{n-2}$. Then

$$\det(B) = \det(Q^t A Q) = \text{tr}_{n-2}(Q Q^t A Q Q^t).$$

Here tr_k means the k -trace; that is, the sum of all the $k \times k$ principal minors for an arbitrary matrix presentation.

Let C^* denote the adjugate matrix of an $(n-1) \times (n-1)$ matrix C . It is simple to see that

$$\text{tr}_{n-2} C = \text{tr} C^*,$$

where $\text{tr} C^*$ is the classical trace of C^* . Meanwhile, a direct computation yields $(Q Q^t)^* = u u^t$. As a result,

$$\begin{aligned} \det(B) &= \det(Q^t A Q) \\ &= \text{tr}_{n-2}(Q Q^t A Q Q^t) \\ &= \text{tr}((Q Q^t)^* A^* (Q Q^t)^*) \\ &= \text{tr}(u u^t A^* u u^t) \\ &= (u^t A^* u) \cdot \text{tr}(u u^t) \\ &= \frac{u_1^2 \kappa_1 + \dots + u_{n-1}^2 \kappa_{n-1}}{\kappa_1 \cdots \kappa_{n-1}}. \end{aligned}$$

This concludes the proof because of (4.1). □

Recall that for compact convex sets $K_1, \dots, K_m$ in $\mathbb{R}^n$ and $\lambda_1, \dots, \lambda_m > 0$, we have

$$V(\lambda_1 K_1 + \dots + \lambda_m K_m) = \sum_{i_1, \dots, i_n=1}^m \lambda_{i_1} \cdots \lambda_{i_n} V(K_{i_1}, \dots, K_{i_n}).$$

where $V(K_{i_1}, \dots, K_{i_n})$ is known as the *mixed volume*, which is nonnegative, symmetric, and multi-linear in each argument. The mixed volume has the following integral representation

$$V(K_1, \dots, K_n) = \frac{1}{n} \int_{S^{n-1}} h_{K_1}(v) dS(K_2, \dots, K_n; v), \quad (4.2)$$

where $S(K_2, \dots, K_n; \cdot)$ is known as the *mixed area measure*.

In comparison, suppose $A_1, \dots, A_m$ are $n \times n$ matrices and $\lambda_1, \dots, \lambda_m > 0$. Then

$$\det(\lambda_1 A_1 + \dots + \lambda_m A_m) = \sum_{i_1, \dots, i_n=1}^m \lambda_{i_1} \cdots \lambda_{i_n} D(A_{i_1}, \dots, A_{i_n}), \quad (4.3)$$

where $D(A_1, \dots, A_n)$ is known as the *mixed discriminant*, which is nonnegative on the set of positive semi-definite matrices, symmetric, and multi-linear in each argument.

When $K_2, \dots, K_n$ have C^2 boundaries with everywhere positive Gauss curvature, the mixed area measure and the mixed discriminant are related in the following way:

$$dS(K_2, \dots, K_n; v) = D(D^2 h_{K_2}(v)|_{v^\perp}, \dots, D^2 h_{K_n}(v)|_{v^\perp}) dv. \quad (4.4)$$

When $f_1, \dots, f_n$ are C^2 functions, and consequently can be written as differences of support functions (see [63, Lemma 1.7.8]), both mixed volume and mixed area measure can be defined:

$$V(f_1, \dots, f_n), \quad S(f_1, \dots, f_{n-1}; \cdot).$$

These functionals are still symmetric, and multi-linear.

For mixed volume and mixed area measures of convex bodies, refer to [63, Section 5.1]; for their extensions to differences of support functions, refer to [63, Section 5.2].

For simplicity, we will write

$$S_{n-2}(K, L; \cdot) = S(K, K, \dots, K, L; \cdot),$$

where K appears $(n-2)$ times. In particular, when L is the centered unit ball, $S_{n-2}(K, B_n; \cdot)$ recovers the $(n-2)$ -area measure $S_{n-2}(K, \cdot)$.

The following corollary is immediate by combining Lemma 4.1 and [66, Lemma 2.1]. For completeness, we provide a short proof.

Corollary 4.2. *Suppose $K \in \mathcal{K}^n$ has C^2 boundary and everywhere positive Gauss curvature. Let $u \in S^{n-1}$. Then,*

$$S_{n-2}(K, [-u, u]; \omega) = \frac{2}{n-1} \int_{\omega \cap u^\perp} \frac{\kappa_K(z, u)}{\kappa_K(z)} d\mathcal{H}^{n-2}(v),$$

where $z = z(v) \in \partial K$ is the boundary point whose normal vector is v .

Proof. By Cauchy projection formula (see, e.g., [63, Theorem 5.3.1]), for all convex body L , we have

$$\frac{1}{2} \int_{S^{n-1}} |u \cdot v| dS_{n-2}(K, L; v) = v(K|u^\perp, K|u^\perp, \dots, K|u^\perp, L|u^\perp),$$

where we use $v(K|u^\perp, K|u^\perp, \dots, K|u^\perp, L|u^\perp)$ for the mixed volume in dimension $(n-1)$ and $K|u^\perp$ appears $(n-2)$ times. This, and equation (4.2) in dimension $(n-1)$, implies

$$\frac{1}{n-1} \int_{S^{n-1} \cap u^\perp} h_L(v) dS_{n-2}(K|u^\perp, v) = \frac{1}{2} \int_{S^{n-1}} h_L(v) dS_{n-2}(K, [-u, u]; v). \quad (4.5)$$

Since the linear span of the class of support functions is dense in $C(S^{n-1})$, we infer that

$$S_{n-2}(K, [-u, u]; \cdot) = \frac{2}{n-1} S_{n-2}(K|u^\perp, \cdot).$$

Now, by Lemma 4.1, we complete the proof. $\square$

Let $K \in \mathcal{K}^n$. Let $\mathbb{P}K$ be the zonoid whose support function is given by

$$h_{\mathbb{P}K}(v) = \int_{S^{n-1}} |v \cdot u| \rho_{\Pi^* K}^{n+1}(u) du, \quad \text{for all } v \in S^{n-1}. \quad (4.6)$$

Coincidentally, $\mathbb{P}K = (n+1)V(K)\Gamma\Pi^*K$, a properly normalized version of the centroid body of the polar projection body of K . See [63, Section 10.8] for the definition of centroid body. It is worth pointing out that for every $\phi \in \text{SL}(n)$, we have $\mathbb{P}\phi K = \phi\mathbb{P}K$.

Definition 4.3. *Let K be a convex body in $\mathbb{R}^n$. Define its affine area measure, $S_{n-2}^{(a)}(K, \cdot)$, to be the finite Borel measure on S^{n-1} given by*

$$S_{n-2}^{(a)}(K, \omega) = \frac{n-1}{4n} S_{n-2}(K, \mathbb{P}K; \omega), \quad \text{for each Borel subset } \omega \subset S^{n-1}.$$

It is simple to see that if $K_i \in \mathcal{K}^n$ converges to $K \in \mathcal{K}^n$ in the Hausdorff metric, then $S_{n-2}^{(a)}(K_i, \cdot)$ converges to $S_{n-2}^{(a)}(K, \cdot)$ weakly on S^{n-1} .

We shall see that the affine area measure $S_{n-2}^{(a)}(K, \cdot)$ naturally arises from differentiating the classical integral affine surface area.

When K has additional regularity, affine area measure is connected to a weighted version of the normal curvature. This can be seen from the following theorem.

Theorem 4.4. *Suppose $K \in \mathcal{K}^n$ has C^2 boundary and everywhere positive Gauss curvature. Then, for each Borel subset $\omega \subset S^{n-1}$, we have*

$$S_{n-2}^{(a)}(K, \omega) = \frac{1}{2n} \int_{\nu_K^{-1}(\omega)} \int_{S^{n-1} \cap \nu_K(z)^\perp} \rho_{\Pi^*K}^{n+1}(u) \kappa_K(z, u) d\mathcal{H}^{n-2}(u) d\mathcal{H}^{n-1}(z).$$

Proof. By Definition 4.3,

$$S_{n-2}^{(a)}(K, \omega) = \frac{n-1}{4n} S_{n-2}(K, \mathbb{P}K; \omega).$$

Note that by the definition of $\mathbb{P}K$ (see (4.6)), the body $\mathbb{P}K$ is a zonoid and consequently the limit of a sequence of zonotopes. Since the mixed area measure $S_{n-2}(K, L; \cdot)$ is linear and continuous in L , we have

$$S_{n-2}^{(a)}(K, \omega) = \frac{n-1}{4n} \int_{S^{n-1}} S_{n-2}(K, [-u, u]; \omega) \rho_{\Pi^*K}^{n+1}(u) du.$$

By Corollary 4.2 and the fact that Radon transform is self-adjoint, we have

$$\begin{aligned} S_{n-2}^{(a)}(K, \omega) &= \frac{1}{2n} \int_{S^{n-1}} \int_{S^{n-1} \cap u^\perp} 1_\omega(v) \frac{\kappa_K(z, u)}{\kappa_K(z)} d\mathcal{H}^{n-2}(v) \rho_{\Pi^*K}^{n+1}(u) du \\ &= \frac{1}{2n} \int_{S^{n-1}} 1_\omega(v) \frac{1}{\kappa_K(z)} \int_{S^{n-1} \cap v^\perp} \kappa_K(z, u) \rho_{\Pi^*K}^{n+1}(u) d\mathcal{H}^{n-2}(u) dv \\ &= \frac{1}{2n} \int_{\nu_K^{-1}(\omega)} \int_{S^{n-1} \cap v^\perp} \kappa_K(z, u) \rho_{\Pi^*K}^{n+1}(u) d\mathcal{H}^{n-2}(u) d\mathcal{H}^{n-1}(z). \end{aligned}$$

Here, the point $z = z(v) \in \partial K$ is the boundary point whose normal vector is v . $\square$

We now show that under additional regularity assumptions on K , we may differentiate the classical integral affine surface area. Note that the derivative here is two-sided.

Theorem 4.5. *Suppose $K \in \mathcal{K}^n$ has C^2 boundary and everywhere positive Gauss curvature. Let $g \in C^2(S^{n-1})$, $h_t = h_K + tg$, and K_t be the Wulff shape generated by h_t . Then,*

$$\left. \frac{d}{dt} \right|_{t=0} \Phi(K_t) = 2\Phi(K)^{n+1} \int_{S^{n-1}} g(v) dS_{n-2}^{(a)}(K, v).$$

Proof. Since $g \in C^2(S^{n-1})$, it can be written as the difference of two support functions.

Since K has C^2 boundary and everywhere positive Gauss curvature, for sufficiently small t , the support function of K_t is exactly h_t . Therefore,

$$\text{vol}_{n-1}(K_t|_{u^\perp}) = \frac{1}{2} \int_{S^{n-1}} |u \cdot v| \det(D^2 h_t(v)) dv.$$

Then, by (4.3), (4.2), (4.4) and their functional extensions, the symmetry of mixed volume, and finally Corollary 4.2, we have

$$\begin{aligned}
& \left. \frac{d}{dt} \right|_{t=0} \text{vol}_{n-1}(K_t|_{u^\perp}) \\
&= \frac{n-1}{2} \int_{S^{n-1}} |u \cdot v| D(D^2 h(v)|_{v^\perp}, \dots, D^2 h(v)|_{v^\perp}, D^2 g(v)|_{v^\perp}) dv \\
&= \frac{(n-1)n}{2} V([-u, u], K, \dots, K, g) \\
&= \frac{(n-1)n}{2} V(K, \dots, K, [-u, u], g) \\
&= \frac{n-1}{2} \int_{S^{n-1}} g(v) dS_{n-2}(K, [-u, u]; v) \\
&= \int_{S^{n-1} \cap u^\perp} g(v) \frac{\kappa_K(z, u)}{\kappa_K(z)} d\mathcal{H}^{n-2}(v),
\end{aligned}$$

where z is the boundary point of K with outer unit normal v . The fact that the Radon transform is self-adjoint now implies

$$\begin{aligned}
\left. \frac{d}{dt} \right|_{t=0} (\Phi(K_t)^{-n}) &= - \int_{S^{n-1}} \rho_{\Pi^* K}^{n+1}(u) \int_{S^{n-1} \cap u^\perp} g(v) \frac{\kappa_K(z, u)}{\kappa_K(z)} d\mathcal{H}^{n-2}(v) du \\
&= - \int_{S^{n-1}} g(v) \frac{1}{\kappa_K(z)} \int_{S^{n-1} \cap v^\perp} \rho_{\Pi^* K}^{n+1}(u) \kappa_K(z, u) d\mathcal{H}^{n-2}(u) dv \\
&= - \int_{\partial K} g(\nu_K(z)) \int_{S^{n-1} \cap \nu_K(z)^\perp} \rho_{\Pi^* K}^{n+1}(u) \kappa_K(z, u) d\mathcal{H}^{n-2}(u) d\mathcal{H}^{n-1}(z) \\
&= - 2n \int_{S^{n-1}} g(v) dS_{n-2}^{(a)}(K, v).
\end{aligned}$$

The desired result follows from the chain rule. $\square$

Without any additional regularity assumptions, in general, only one-sided derivative is possible. (Recall that the situation is similar for surface area.)

Theorem 4.6. *Let K be a convex body and L be a compact convex set in $\mathbb{R}^n$. Then,*

$$\left. \frac{d}{dt} \right|_{t=0^+} \Phi(K + tL) = 2\Phi(K)^{n+1} \int_{S^{n-1}} h_L(v) dS_{n-2}^{(a)}(K, v).$$

Proof. By the (classical) Aleksandrov variational formula for the volume functional and (4.5), we have

$$\begin{aligned}
\left. \frac{d}{dt} \right|_{t=0^+} \text{vol}_{n-1}((K + tL)|_{u^\perp}) &= \left. \frac{d}{dt} \right|_{t=0} \text{vol}_{n-1}(K|_{u^\perp} + tL|_{u^\perp}) \\
&= (n-1)v(K|_{u^\perp}, K|_{u^\perp}, \dots, K|_{u^\perp}, L|_{u^\perp}) \\
&= \frac{n-1}{2} \int_{S^{n-1}} h_L(v) dS_{n-2}(K, [-u, u]; v),
\end{aligned}$$

where as before, we use $v(K|u^\perp, K|u^\perp, \dots, K|u^\perp, L|u^\perp)$ to denote the mixed volume in $u^\perp$. Hence, by the fact that the mixed area measure $S_{n-2}(K, L; \cdot)$ is linear and continuous in L , we have

$$\begin{aligned} \left. \frac{d}{dt} \right|_{t=0^+} (\Phi(K + tL)^{-n}) &= -\frac{n-1}{2} \int_{S^{n-1}} \text{vol}_{n-1}(K|u^\perp)^{-n-1} \int_{S^{n-1}} h_L(v) dS_{n-2}(K, [-u, u]; v) du \\ &= -\frac{n-1}{2} \int_{S^{n-1}} h_L(v) dS_{n-2}(K, \mathbb{P}K; v) \\ &= -2n \int_{S^{n-1}} h_L(v) dS_{n-2}^{(a)}(K, v). \end{aligned}$$

The desired result immediately follows from the chain rule. $\square$

The following Brunn-Minkowski inequality for integral affine surface area is due to Lutwak [47]. Since equality conditions were not discussed there and they are critical for Theorem 4.8, we provide a quick proof of the inequality along with the equality conditions.

Theorem 4.7. *Let $K, L \in \mathcal{K}^n$ and $\lambda \in (0, 1)$. Denote $K_\lambda = (1 - \lambda)K + \lambda L$. Then*

$$\Phi(K_\lambda) \geq \Phi(K)^{1-\lambda} \Phi(L)^\lambda, \quad (4.7)$$

with equality if and only if K and L are translations of each other.

Proof. Let $u \in S^{n-1}$ be arbitrarily fixed. By the Brunn-Minkowski inequality, we have

$$\text{vol}_{n-1}(K_\lambda|u^\perp) \geq \text{vol}_{n-1}(K|u^\perp)^{1-\lambda} \text{vol}_{n-1}(L|u^\perp)^\lambda, \quad (4.8)$$

with equality if and only if $K|u^\perp$ and $L|u^\perp$ are translations of each other. Hence,

$$\begin{aligned} \Phi(K_\lambda)^{-n} &= \frac{1}{n} \int_{S^{n-1}} \text{vol}_{n-1}(K_\lambda|u^\perp)^{-n} du \\ &\leq \frac{1}{n} \int_{S^{n-1}} (\text{vol}_{n-1}(K|u^\perp)^{1-\lambda} \text{vol}_{n-1}(L|u^\perp)^\lambda)^{-n} du \\ &\leq \left(\frac{1}{n} \int_{S^{n-1}} \text{vol}_{n-1}(K|u^\perp)^{-n} du \right)^{1-\lambda} \left(\frac{1}{n} \int_{S^{n-1}} \text{vol}_{n-1}(L|u^\perp)^{-n} du \right)^\lambda \\ &= \Phi(K)^{-n(1-\lambda)} \Phi(L)^{-n\lambda}. \end{aligned}$$

Here, Hölder's inequality is used to derive the second inequality.

If equality holds in (4.7), then equality must hold in (4.8) for all $u \in S^{n-1}$. Thus, for every $u \in S^{n-1}$,

$$\text{vol}_{n-1}(K_\lambda|u^\perp) = \text{vol}_{n-1}(K|u^\perp) = \text{vol}_{n-1}(L|u^\perp).$$

Integrating with respect to u , we get

$$S(K_\lambda) = S(K)^{1-\lambda} S(L)^\lambda.$$

This, in particular, implies that equality holds in the Brunn-Minkowski inequality for surface area. Consequently, K and L are translations of each other. The Brunn-Minkowski inequality for surface area can be found in [63, Theorem 7.4.5], and its equality condition can be seen by combining Theorem 7.4.6 and Corollary 8.1.4 there. $\square$

Since $\Phi(K)$ is homogeneous of degree $(n-1)$ in K , it is immediate that Theorem 4.7 is equivalent to the following additive form of the Brunn-Minkowski inequality:

$$\Phi(K+L)^{\frac{1}{n-1}} \geq \Phi(K)^{\frac{1}{n-1}} + \Phi(L)^{\frac{1}{n-1}}, \quad (4.9)$$

with equality if and only if K and L are homothetic.

The next theorem reveals that the affine area measure contains enough information to uniquely determine the shape of a convex body.

Theorem 4.8. *Let K, L be two convex bodies in $\mathbb{R}^n$. If $S_{n-2}^{(a)}(K, \cdot) = S_{n-2}^{(a)}(L, \cdot)$, then K and L are translations of each other.*

Proof. Apply Theorem 4.6 with $K_t = K + tK$, we get

$$\left. \frac{d}{dt} \right|_{t=0+} \Phi(K + tK) = 2\Phi(K)^{n+1} \int_{S^{n-1}} h_K(v) dS_{n-2}^{(a)}(K, v).$$

On the other hand, since $\Phi((1+t)K) = (1+t)^{n-1}\Phi(K)$, we have

$$\left. \frac{d}{dt} \right|_{t=0+} \Phi(K + tK) = (n-1)\Phi(K).$$

Consequently, we have

$$\frac{2}{n-1} \int_{S^{n-1}} h_K(v) dS_{n-2}^{(a)}(K, v) = \Phi(K)^{-n}. \quad (4.10)$$

By (4.9), we have

$$\Phi(K + tL)^{\frac{1}{n-1}} \geq \Phi(K)^{\frac{1}{n-1}} + t\Phi(L)^{\frac{1}{n-1}},$$

with equality if and only if K and L are homothetic. Using Theorem 4.6, we have

$$\frac{2}{n-1} \Phi(K)^{\frac{1}{n-1}+n} \int_{S^{n-1}} h_L(v) dS_{n-2}^{(a)}(K, v) \geq \Phi(L)^{\frac{1}{n-1}}. \quad (4.11)$$

Note that equality holds in (4.11) if and only if K and L are homothetic. This follows in the same spirit as the equivalence of the classical Brunn-Minkowski inequality and the classical Minkowski inequality. See, for example, [63, Theorem 7.2.1]. Here, we provide a quick sketch. Indeed, Theorem 4.7 (equivalently, (4.9)) and the homogeneity of Φ imply that the function

$$f(t) = \Phi(K + tL)^{\frac{1}{n-1}} - \Phi(K)^{\frac{1}{n-1}} - t\Phi(L)^{\frac{1}{n-1}},$$

is concave in t . Moreover, $f(0) = 0$ and $f(t) \geq 0$ for all $t \geq 0$. If equality holds in (4.11), then $f'(0) = 0$. By the concavity and non-negativity of f , we have $f(t) = 0$ for all $t \geq 0$. In particular, $f(1) = 0$, which implies that equality holds in (4.9).

Since $S_{n-2}^{(a)}(K, \cdot) = S_{n-2}^{(a)}(L, \cdot)$, by (4.10), we have

$$\begin{aligned} \frac{2}{n-1} \Phi(K)^{\frac{1}{n-1}+n} \int_{S^{n-1}} h_L(v) dS_{n-2}^{(a)}(K, v) &= \frac{2}{n-1} \Phi(K)^{\frac{1}{n-1}+n} \int_{S^{n-1}} h_L(v) dS_{n-2}^{(a)}(L, v) \\ &= \Phi(K)^{\frac{1}{n-1}+n} \Phi(L)^{-n}, \end{aligned}$$

which, when combined with (4.11), implies,

$$\Phi(K) \geq \Phi(L).$$

Reversing the role of K and L , one immediately get $\Phi(K) = \Phi(L)$. Consequently, equality holds in (4.11). The equality condition of (4.11) implies that K and L are homothetic. Finally, the fact that $\Phi(K) = \Phi(L)$ implies that K and L are translations of each other. $\square$

Recall that as $q \rightarrow -1^+$, the fractional integral affine surface area $\Phi_q(K)$ converges to the classical integral affine surface area $\Phi(K)$ of K . We show that under sufficient regularity assumptions, the limit of $F_q^{(a)}(K, \cdot)$, as $q \rightarrow -1^+$, is precisely $S_{n-2}^{(a)}(K, \cdot)$.

The following lemma is needed.

Lemma 4.9. *Let K be a convex body in $\mathbb{R}^n$ with C^2 boundary and $z \in \partial K$. Then,*

$$\lim_{q \rightarrow -1^+} (q+1) \int_{S^{n-1}} \rho_{S_q K}^{n-q}(u) X_K^q(z, u) du = \int_{\nu_K(z)^\perp \cap S^{n-1}} \rho_{\Pi^* K}^{n+1}(\theta) \kappa_K(z, \theta) d\mathcal{H}^{n-2}(\theta).$$

Proof. Let $\varepsilon > 0$ be arbitrary.

Since everything is translation invariant, we may assume $z = o$. Moreover, since K has C^2 boundary, we may set up a coordinate system $(o; e_1, \dots, e_n)$ such that $e_n = -\nu_K(z)$ and in a small neighborhood of o , the boundary ∂K can locally be represented as the graph of the function

$$y_n = f(y') = \frac{1}{2}(\kappa_1 y_1^2 + \dots + \kappa_{n-1} y_{n-1}^2) + o(|y'|^2). \quad (4.12)$$

Here $y = (y', y_n) = (y_1, \dots, y_{n-1}, y_n)$ is the coordinate with respect to $(o; e_1, \dots, e_n)$ and κ_i are the principal curvatures of K at the boundary point o .

It is also worth noting that since ∂K is C^2 , we have the following identity:

$$(q+1) \int_{S^{n-1}} \rho_{S_q K}^{n-q}(u) X_K^q(o, u) du = 2(q+1) \int_{\{u \in S^{n-1} : u \cdot e_n > 0\}} \rho_{S_q K}^{n-q}(u) \rho_K^q(u) du. \quad (4.13)$$

Note also that in the neighborhood where (4.12) represents ∂K locally, we have the following identities:

$$\begin{aligned} \nu_K(y) &= \frac{(\nabla f(y'), -1)}{\sqrt{1 + |\nabla f(y')|^2}}, \\ d\mathcal{H}^{n-1}(y) &= \sqrt{1 + |\nabla f(y')|^2} d\mathcal{H}^{n-1}(y'). \end{aligned} \quad (4.14)$$

We write in polar coordinates: $0 \neq y' = r\theta$ where $r > 0$ and $\theta \in S^{n-2} = S^{n-1} \cap e_n^\perp$. Note that

$$\begin{aligned} \left| \frac{(r\theta, f(r\theta))}{\sqrt{r^2 + f(r\theta)^2}} - (\theta, 0) \right| &= \left| \frac{(\theta, f(r\theta)/r)}{\sqrt{1 + f(r\theta)^2/r^2}} - (\theta, 0) \right| \\ &= \frac{1}{\sqrt{1 + f(r\theta)^2/r^2}} \left| \left(\left(1 - \sqrt{1 + \frac{f^2(r\theta)}{r^2}} \right) \theta, \frac{f(r\theta)}{r} \right) \right| \\ &= \frac{1}{\sqrt{1 + f(r\theta)^2/r^2}} \left[\left(1 - \sqrt{1 + \frac{f^2(r\theta)}{r^2}} \right)^2 + \frac{f^2(r\theta)}{r^2} \right]^{\frac{1}{2}}. \end{aligned}$$

Note that $f(r\theta) = O(r^2)$ when $r > 0$ is sufficiently small. This and the fact that $\rho_{\Pi^* K}$ is uniformly continuous on S^{n-1} , imply the existence of $\delta > 0$ such that for $r \in (0, \delta)$, we have

$$\left| \rho_{\Pi^* K}^{n+1} \left(\frac{(r\theta, f(r\theta))}{\sqrt{r^2 + f(r\theta)^2}} \right) - \rho_{\Pi^* K}^{n+1}((\theta, 0)) \right| < \varepsilon. \quad (4.15)$$

Since $\rho_{S_q K}$ converges to $\rho_{\Pi^* K}$ uniformly on S^{n-1} as $q \rightarrow -1^+$ and we are only interested in the limiting behavior, we may restrict ourselves to those $q > -1$ such that

$$|\rho_{S_q K}^{n-q} - \rho_{\Pi^* K}^{n+1}| < \varepsilon, \quad \text{on } S^{n-1}. \quad (4.16)$$

In a small punctured neighborhood of o , we have the following relationship between u and y :

$$\rho_K(u)u = y = (y', f(y')). \quad (4.17)$$

Moreover,

$$\rho_K(u)^n du = |y \cdot \nu_K(y)| d\mathcal{H}^{n-1}(y), \quad y' \cdot \nabla f(y') = 2f(y') + o(|y'|^2).$$

With (4.14), we have

$$\rho_K(u)^n du = (f(y') + o(|y'|^2)) d\mathcal{H}^{n-1}(y'). \quad (4.18)$$

We are finally ready to compute the limit of the right-hand side of (4.13). We do so by separating the integral domain into

$$\begin{aligned} D_1 &= \{u \in S^{n-1} : u \cdot e_n > 0, |y'| \in (0, \delta)\}, \\ D_2 &= \{u \in S^{n-1} : u \cdot e_n > 0\} \setminus D_1, \end{aligned}$$

where $\delta > 0$ is sufficiently small so that for all $u \in D_1$, the boundary point $u\rho_K(u) = (y', y_n) \in \partial K$ is on the graph of f given in (4.12).

We first focus on the integration on D_1 . By (4.16), (4.18), (4.17),

$$\begin{aligned} & (q+1) \int_{D_1} \rho_{S_q K}^{n-q}(u) \rho_K^q(u) du \\ & \leq (q+1) \int_{D_1} (\rho_{\Pi^* K}^{n+1}(u) + \varepsilon) \rho_K^q(u) du \\ & = (q+1) \int_{|y'| \in (0, \delta)} \left(\rho_{\Pi^* K}^{n+1} \left(\frac{(y', f(y'))}{|(y', f(y'))|} \right) + \varepsilon \right) |(y', f(y'))|^{q-n} (f(y') + o(|y'|^2)) d\mathcal{H}^{n-1}(y') \\ & = (q+1) \int_{S^{n-2}} \int_0^\delta \left(\rho_{\Pi^* K}^{n+1} \left(\frac{(r\theta, f(r\theta))}{|(r\theta, f(r\theta))|} \right) + \varepsilon \right) |(r\theta, f(r\theta))|^{q-n} (f(r\theta) + o(r^2)) r^{n-2} dr d\mathcal{H}^{n-2}(\theta). \end{aligned}$$

Recall the normal curvature of K at o in the direction θ is given by $\kappa_K(o, \theta) = \kappa_1(\theta \cdot e_1)^2 + \dots + \kappa_{n-1}(\theta \cdot e_{n-1})^2$. Thus, $f(r\theta) = \frac{1}{2}r^2 \kappa_K(o, \theta) + o(r^2)$. Therefore,

$$\begin{aligned} & (q+1) \int_{D_1} \rho_{S_q K}^{n-q}(u) \rho_K^q(u) du \\ & \leq (q+1) \int_{S^{n-2}} \int_0^\delta \left(\rho_{\Pi^* K}^{n+1} \left(\frac{(r\theta, f(r\theta))}{|(r\theta, f(r\theta))|} \right) + \varepsilon \right) r^q \left(1 + \frac{r^2 \kappa_K(o, \theta)^2}{4} + o(r^2) \right)^{\frac{q-n}{2}} \\ & \quad \cdot \left(\frac{\kappa_K(o, \theta)}{2} + o(1) \right) dr d\mathcal{H}^{n-2}(\theta) \\ & \leq (q+1) \int_{S^{n-2}} \int_0^\delta \left(\rho_{\Pi^* K}^{n+1} \left(\frac{(r\theta, f(r\theta))}{|(r\theta, f(r\theta))|} \right) + \varepsilon \right) r^q \frac{\kappa_K(o, \theta)}{2} dr d\mathcal{H}^{n-2}(\theta) + (q+1)o(1) \int_0^\delta r^q dr, \end{aligned}$$

Here, in the last step, we used the fact that all terms in the integrand except for r^q are uniformly bounded from above and from below away from 0. Now, using (4.15), we have

$$\begin{aligned} & (q+1) \int_{D_1} \rho_{S_q K}^{n-q}(u) \rho_K^q(u) du \\ & \leq (q+1) \int_{S^{n-2}} \int_0^\delta (\rho_{\Pi^* K}^{n+1}(\theta, 0) + 2\varepsilon) r^q \frac{\kappa_K(o, \theta)}{2} dr d\mathcal{H}^{n-2}(\theta) + o(1)\delta^{q+1} \\ & = \frac{1}{2} \int_{S^{n-2}} (\rho_{\Pi^* K}^{n+1}(\theta, 0) + 2\varepsilon) \kappa_K(o, \theta) d\mathcal{H}^{n-2}(\theta) \delta^{q+1} + o(1)\delta^{q+1}. \end{aligned}$$

Let $q \rightarrow -1^+$. Since $\varepsilon > 0$ is arbitrary, we have

$$\limsup_{q \rightarrow -1^+} (q+1) \int_{D_1} \rho_{S_q K}^{n-q}(u) \rho_K^q(u) du \leq \frac{1}{2} \int_{S^{n-2}} \rho_{\Pi^* K}^{n+1}(\theta, 0) \kappa_K(o, \theta) d\mathcal{H}^{n-2}(\theta) + o(1). \quad (4.19)$$

It is simple to see that for $u \in D_2$, we have $\rho_K(u) \geq \delta$. This, when combined with the fact that $\rho_{S_q K}(u)$ is uniformly bounded in q and u and (4.19), implies that

$$\limsup_{q \rightarrow -1^+} (q+1) \int_{\{u \in S^{n-1} : u \cdot e_n > 0\}} \rho_{S_q K}^{n-q}(u) \rho_K^q(u) du \leq \frac{1}{2} \int_{S^{n-2}} \rho_{\Pi^* K}^{n+1}(\theta, 0) \kappa_K(o, \theta) d\mathcal{H}^{n-2}(\theta) + o(1).$$

Sending $\delta \rightarrow 0^+$ now yields

$$\limsup_{q \rightarrow -1^+} (q+1) \int_{\{u \in S^{n-1} : u \cdot e_n > 0\}} \rho_{S_q K}^{n-q}(u) \rho_K^q(u) du \leq \frac{1}{2} \int_{S^{n-2}} \rho_{\Pi^* K}^{n+1}(\theta, 0) \kappa_K(o, \theta) d\mathcal{H}^{n-2}(\theta). \quad (4.20)$$

Repeating the same arguments (except this time, when using (4.15) and (4.16), we use the other side of the bound), we get

$$\liminf_{q \rightarrow -1^+} (q+1) \int_{\{u \in S^{n-1} : u \cdot e_n > 0\}} \rho_{S_q K}^{n-q}(u) \rho_K^q(u) du \geq \frac{1}{2} \int_{S^{n-2}} \rho_{\Pi^* K}^{n+1}(\theta, 0) \kappa_K(o, \theta) d\mathcal{H}^{n-2}(\theta). \quad (4.21)$$

The desired result now follows from (4.13), (4.20), and (4.21). $\square$

Theorem 4.10. *Let K be a convex body in $\mathbb{R}^n$ with C^2 boundary with everywhere positive Gauss curvature. Then, for each Borel set $\omega \subset S^{n-1}$, we have*

$$\lim_{q \rightarrow -1^+} F_q^{(a)}(K, \omega) = S_{n-2}^{(a)}(K, \omega).$$

Proof. Since K has non-empty interior and $\rho_{S_q K}^{n-q}$ converges uniformly to $\rho_{\Pi^* K}^{n+1}$ on S^{n-1} as $q \rightarrow -1^+$, there exists $C > 0$ such that for every $z \in \partial K$,

$$(q+1) \int_{S^{n-1}} \rho_{S_q K}^{n-q}(u) X_K^q(z, u) du \leq C(q+1) \int_{S^{n-1}} X_K^q(z, u) du = 2nC(q+1) \tilde{V}_q(K, z).$$

It was shown in [51, Corollary 4.14] that

$$\lim_{q \rightarrow -1^+} \int_{\nu_K^{-1}(\omega)} (q+1) \tilde{V}_q(K, z) d\mathcal{H}^{n-1}(z) = \int_{\nu_K^{-1}(\omega)} \lim_{q \rightarrow -1^+} (q+1) \tilde{V}_q(K, z) d\mathcal{H}^{n-1}(z).$$

Thus, by using the generalized dominated convergence theorem, we have

$$\begin{aligned} \lim_{q \rightarrow -1^+} F_q^{(a)}(K, \omega) &= \frac{1}{2n} \lim_{q \rightarrow -1^+} \int_{\nu_K^{-1}(\omega)} (q+1) \int_{S^{n-1}} \rho_{S_q K}^{n-q}(u) X_K^q(z, u) du d\mathcal{H}^{n-1}(z) \\ &= \frac{1}{2n} \int_{\nu_K^{-1}(\omega)} \lim_{q \rightarrow -1^+} \left((q+1) \int_{S^{n-1}} \rho_{S_q K}^{n-q}(u) X_K^q(z, u) du \right) d\mathcal{H}^{n-1}(z). \end{aligned}$$

The desired result now follows from Lemma 4.9 and the definition of the affine area measure. $\square$

Remark 4.11. *At this point, we remark that there are three separate (yet equivalent) ways that we could introduce the affine area measure $S_{n-2}^{(a)}(K, \cdot)$:*

- (1) *define $S_{n-2}^{(a)}(K, \cdot)$ through the one-sided derivative of $\Phi(K + tL)$ at $t = 0^+$ as computed in Theorem 4.6;*
- (2) *define $S_{n-2}^{(a)}(K, \cdot)$ for those K with C^2 boundary and everywhere positive Gauss curvature through the double-sided derivative as computed in Theorem 4.5 and for generic convex bodies, we take the weak limit of $S_{n-2}^{(a)}(K_i, \cdot)$ where K_i is a sequence of smooth convex bodies converging to an arbitrarily fixed generic convex body K ;*
- (3) *define $S_{n-2}^{(a)}(K, \cdot)$ for those K with C^2 boundary and everywhere positive Gauss curvature through the limit of $F_q^{(a)}(K, \cdot)$ as $q \rightarrow -1^+$ as established in Theorem 4.10 and for generic convex bodies, we approximate as outlined in (2).*

5. MINKOWSKI PROBLEMS FOR FRACTIONAL AFFINE AREA MEASURES

The characterization problem for the classical surface area measure $S_{n-1}(K, \cdot)$, known as the classical Minkowski problem, was investigated throughout the last century and has served as a major motivation for Monge-Ampère equations. Motivated by this, it is natural to ask the following Minkowski-type problem.

The L_p Minkowski problem for fractional affine area measure. Let $0 \neq q > 1$, $p \in \mathbb{R}$, and μ be a finite Borel measure on S^{n-1} . What are the necessary and sufficient condition(s) on μ so that there exists a convex body K in $\mathbb{R}^n$ (containing the origin in their interiors if $p \neq 1$) such that

$$\mu = F_{p,q}^{(a)}(K, \cdot)? \quad (5.1)$$

When $\mu = f dv$ is absolutely continuous, the measure equation (5.1) reduces to the following nonlocal (and rather complicated) Monge-Ampère equation on S^{n-1} :

$$\tilde{V}_q([h], S_q[h], \nabla_{S^{n-1}} h(v) + h(v)v) h^{1-p}(v) \det(\nabla_{S^{n-1}}^2 h(v) + h(v)I) = f, \quad (5.2)$$

where one is searching for a convex solution $h \in C^2(S^{n-1})$ such that h is the support function of the sought-for K in (5.1).

Accompanied by the measure equation (5.1) is the uniqueness and regularity of solutions, which we state but will not address in the current work.

Uniqueness of solutions to the L_p Minkowski problem for fractional affine area measure. If K and L are convex bodies in $\mathbb{R}^n$ (containing the origin in its interior unless $p = 1$) and

$$F_{p,q}^{(a)}(K, \cdot) = F_{p,q}^{(a)}(L, \cdot),$$

to what extent is K related to L ?

A satisfactory answer to the uniqueness question (in the affirmative direction) will show that the fractional affine area measure $F_{p,q}^{(a)}(K, \cdot)$ encodes all information regarding the shape of K .

Regularity of solutions to the L_p Minkowski problem for fractional affine area measure. If f in (5.2) is $C^{k,\alpha}(S^{n-1})$ for some $k \in \mathbb{N}$ and $\alpha > 0$, what regularity does the solution h have?

While it is the case that the regularity of solutions to the classical Minkowski problem (and many other Minkowski problems since then) follows from the landmark work [11, 12, 13] laid by Caffarelli around the 1990s, this is not the case for (5.2). We emphasize again that (5.2) is *nonlocal*; in fact, to study the regularity of h , one must study the regularity of the maps $h \mapsto \rho_{S_q[h]}$ and $h \mapsto X_{[h]}$, both of which are *highly* nontrivial.

The rest of this section will focus on the existence of solutions to (5.1). In fact, we will only treat the two critical cases $p = 1$ and $p = 0$.

The case $p = 1$ for surface area measure was classically treated by Minkowski, Fenchel-Jessen, Aleksandrov, Pogorelov, and Cheng-Yau, among many others. The L_1 fractional affine area measure is the only one in its family that is translation-invariant.

The case $p = 0$ for surface area measure, more commonly known as *the cone volume measure*, is an important singular case. The name *cone volume measure* is partly due to the fact that the total mass of it is equal to the volume of a convex body and that, in the case of a polytope, the cone volume measure encodes data regarding the volume of the convex hulls of the origin and each facet of the body. It was revealed in the seminal work [8] by Böröczky-Lutwak-Yang-Zhang that in this singular case, measure concentration phenomenon will arise. In this case, the L_0 problem is also more commonly known as the logarithmic problem. As revealed in Theorem 3.15, this is the only case where the measure is $\text{SL}(n)$ -contravariant.

5.1. $p = 1$: the Minkowski problem for fractional affine area measure.

The proof of Theorem 1.2 is the main objective in this subsection. Its proof will be variational and will be immediate after several preparatory lemmas.

Consider the following optimization problem:

$$\inf_{K \in \mathcal{K}^n} \left\{ \int_{S^{n-1}} h_K(v) d\mu(v) : \Phi_q(K) = 1 \right\}. \quad (5.3)$$

That there exists at least one $K \in \mathcal{K}^n$ such that $\Phi_q(K) = 1$ follows from the fact that $\Phi_q(K)$ is homogeneous of degree $(n + q)$ in K .

Lemma 5.1. *Let $0 \neq q > -1$ and μ be a finite Borel measure on S^{n-1} . If $K_0 \in \mathcal{K}^n$ satisfies $\Phi_q(K_0) = 1$ and is a minimizer to (5.3), then there exists $c > 0$ such that*

$$\mu = F_q^{(a)}(cK_0, \cdot).$$

Proof. We claim that h_{K_0} is a minimizer to the following modified optimization problem:

$$\inf_{h \in C(S^{n-1})} \left\{ \int_{S^{n-1}} h(v) d\mu(v) : \Phi_q([h]) = 1, [h] \in \mathcal{K}^n \right\}. \quad (5.4)$$

Indeed, for every $h \in C(S^{n-1})$ with $\Phi_q([h]) = 1$, by the fact that K_0 is a minimizer to (5.3) and the definition of Wulff shape, we conclude that

$$\int_{S^{n-1}} h_{K_0}(v) d\mu(v) \leq \int_{S^{n-1}} h_{[h]}(v) d\mu(v) \leq \int_{S^{n-1}} h(v) d\mu(v).$$

Since h is arbitrary, this proves our claim.

Now, let $f \in C(S^{n-1})$ be arbitrary and consider $h_t = h_{K_0} + tf$, and $K_t = [h_t]$. Since K_0 contains an interior point and $f \in C(S^{n-1})$, when $|t|$ is sufficiently small, K_t is a convex body in $\mathbb{R}^n$ with a non-empty interior. Denote $\gamma(t) = \Phi_q(K_t)^{-\frac{1}{n+q}}$ and consider $\overline{K_t} = \gamma(t)K_t$. By the homogeneity of Φ_q and the choice of $\gamma(t)$, it is simple to see $\Phi_q(\overline{K_t}) = 1$. Moreover, $\gamma(0) = 1$. Since h_{K_0} is a minimizer to (5.4), we have

$$0 = \frac{d}{dt} \Big|_{t=0} \int_{S^{n-1}} \gamma(t) h_t(v) d\mu(v) = \gamma'(0) \int_{S^{n-1}} h_{K_0} d\mu + \int_{S^{n-1}} f d\mu. \quad (5.5)$$

By Theorem 1.1, we have

$$\gamma'(0) = -\frac{2}{n+q} \int_{S^{n-1}} f(v) dF_q^{(a)}(K_0, v). \quad (5.6)$$

Since f is arbitrary, equations (5.5) and (5.6) imply

$$\left(\frac{2}{n+q} \int_{S^{n-1}} h_{K_0} d\mu \right) dF_q^{(a)}(K_0, v) = d\mu.$$

The desired result comes from the homogeneity of $F_q^{(a)}(K, \cdot)$ in K and a proper scaling of K_0 . $\square$

Lemma 5.2. *Let $0 \neq q > -1$. There exists $c > 0$ dependent only on n and q , such that*

$$\frac{1}{c} \leq \frac{\Phi_q^{\frac{1}{n+q}}(K)}{V_n^{\frac{1}{n}}(K)} \leq c$$

for all $K \in \mathcal{K}^n$.

Proof. Let $K_i \in \mathcal{K}^n$ be a minimizing sequence for the ratio

$$\frac{\Phi_q^{\frac{1}{n+q}}(K)}{V_n^{\frac{1}{n}}(K)}. \quad (5.7)$$

Let JK_i be the John ellipsoid of K_i . Since the ratio (5.7) is translation and $GL(n)$ invariant, we may assume that $JK_i = B_n$. By John's theorem, we conclude that

$$B_n \subset K_i \subset nB_n.$$

Since K_i is uniformly bounded, by Blaschke's selection theorem, we may extract a convergent subsequence K_{i_j} that converges in the Hausdorff metric to Q_1 . Since $B_n \subset K_i$, this implies $B_n \subset Q_1$. In particular, $Q_1 \in \mathcal{K}^n$. Since the ratio (5.7) is continuous on $\mathcal{K}^n$ with respect to the Hausdorff metric, we conclude that

$$\frac{\Phi_q^{\frac{1}{n+q}}(Q_1)}{V_n^{\frac{1}{n}}(Q_1)} \leq \frac{\Phi_q^{\frac{1}{n+q}}(K)}{V_n^{\frac{1}{n}}(K)}.$$

A similar argument yields the existence of $Q_2 \in \mathcal{K}^n$ such that

$$\frac{\Phi_q^{\frac{1}{n+q}}(K)}{V_n^{\frac{1}{n}}(K)} \leq \frac{\Phi_q^{\frac{1}{n+q}}(Q_2)}{V_n^{\frac{1}{n}}(Q_2)}.$$

Choosing

$$c = \max \left\{ \frac{\Phi_q^{\frac{1}{n+q}}(Q_2)}{V^{\frac{1}{n}}(Q_2)}, \frac{V^{\frac{1}{n}}(Q_1)}{\Phi_q^{\frac{1}{n+q}}(Q_1)} \right\}$$

settles the proof. $\square$

Lemma 5.3. *Let $0 \neq q > -1$ and μ be a finite Borel measure on S^{n-1} . If μ is not concentrated in any subsphere and*

$$\int_{S^{n-1}} v d\mu(v) = o, \quad (5.8)$$

then there exists $K_0 \in \mathcal{K}^n$ such that $\Phi_q(K_0) = 1$ and K_0 is a minimizer to (5.3).

Proof. Let $K_i \in \mathcal{K}^n$ satisfying $\Phi_q(K_i) = 1$ be a minimizing sequence to (5.3). By (5.8), we have

$$\int_{S^{n-1}} h_{K_i+x} d\mu = \int_{S^{n-1}} h_{K_i} d\mu + \int_{S^{n-1}} x \cdot v d\mu(v) = \int_{S^{n-1}} h_{K_i} d\mu.$$

This, in combination with the fact that Φ_q is translation invariant, allows us to assume without loss of generality that the origin is an interior point of K_i .

Let $M_i = \max_{u \in S^{n-1}} \rho_{K_i}(u) = \rho_{K_i}(u_0)$ for some $u_0 \in S^{n-1}$. By definition of support function, we have

$$\int_{S^{n-1}} h_{K_i} d\mu \geq M_i \int_{S^{n-1}} \max\{u_0 \cdot v, 0\} d\mu(v).$$

Since μ is not concentrated in any subsphere, it follows from (5.8) that it is not concentrated in any closed hemisphere. Therefore, the function

$$u \mapsto \int_{S^{n-1}} \max\{u \cdot v, 0\} d\mu(v)$$

is a positive continuous function on S^{n-1} and consequently has a lower bound $c_0 > 0$ on S^{n-1} . Hence,

$$\int_{S^{n-1}} h_{K_i} d\mu \geq M_i c_0.$$

Since K_i is a minimization sequence, this shows that M_i is uniformly bounded from above. Let $M = \max M_i < \infty$. Consequently $K_i \subset MB_n$. Now, we may use Blaschke's selection theorem to choose a convergent subsequence K_{i_j} that converges in the Hausdorff metric to a convex compact set $K_0 \subset \mathbb{R}^n$.

By Lemma 5.2 and the fact that $\Phi_q(K_i) = 1$, we conclude that $V(K_i) \geq c_1$ for some $c_1 > 0$. By continuity of the volume functional, we conclude that $V(K_0) \geq c_1$, which in turn shows that K_0 has a nonempty interior and consequently $K_0 \in \mathcal{K}^n$. The continuity of Φ_q on $\mathcal{K}^n$ now yields $\Phi_q(K_0) = 1$. This concludes the proof of the lemma. $\square$

We are now in a position to prove Theorem 1.2.

Proof of Theorem 1.2. Suppose first that $\mu = F_q^{(a)}(K, \cdot)$ for some $K \in \mathcal{K}^n$. By translation invariance, we may assume $K \in \mathcal{K}_o^n$. This, in particular, shows the existence of $c_1 > 0$ such that

$$\frac{1}{c_1} B_n \subset K \subset c_1 B_n.$$

The fact that μ is not concentrated in any subsphere now follows from Definition 3.1. Equation (1.6) follows from Theorem 3.16.

Let us now assume μ is not concentrated in any subsphere, and it satisfies (1.6). The existence of K that satisfies $\mu = F_q^{(a)}(K, \cdot)$ follows directly from Lemmas 5.1 and 5.3. $\square$

5.2. $p = 0$: the logarithmic Minkowski problem for fractional affine area measure.

In this section, we provide a proof of Theorem 1.3. The main observation is that since $\Phi_q(K)$ is $\text{SL}(n)$ invariant, a slight modification of the arguments in [8] reveals the involvement of measure concentration condition in the logarithmic problem for fractional affine area measure.

Consider

$$\inf_{K \in \mathcal{K}_e^n} \left\{ \int_{S^{n-1}} \log h_K(v) d\mu(v) : \Phi_q(K) = \left[\frac{2\mu(S^{n-1})}{q+n} \right]^{\frac{q}{n}} \right\}. \quad (5.9)$$

Lemma 5.4. *Let $0 \neq q > -1$ and μ be a finite even Borel measure on S^{n-1} . If $K_0 \in \mathcal{K}_e^n$ satisfies $\Phi_q(K_0) = \left[\frac{2\mu(S^{n-1})}{q+n} \right]^{\frac{q}{n}}$ and is a minimizer to (5.9), then*

$$\mu = F_{0,q}^{(a)}(K_0, \cdot),$$

Proof. Similar to the proof of Lemma 5.1, one may show that h_{K_0} is a minimizer to

$$\inf_{h \in C_e^+(S^{n-1})} \left\{ \int_{S^{n-1}} \log h(v) d\mu(v) : \Phi_q([h]) = \left[\frac{2\mu(S^{n-1})}{q+n} \right]^{\frac{q}{n}} \right\}.$$

Now, let $f \in C_e(S^{n-1})$ be arbitrary and set up $h_t = h e^{tf}$, $K_t = [h_t]$,

$$\gamma(t) = \left(\frac{2\mu(S^{n-1})}{(q+n) \Phi_q(K_t)^{\frac{n}{q}}} \right)^{\frac{q}{n(q+n)}}$$

and $\overline{K_t} = \gamma(t) K_t$. Then,

$$\begin{aligned} & \int_{S^{n-1}} \log(\gamma(t) h_t(v)) d\mu(v) \\ &= \mu(S^{n-1}) \log \gamma(t) + \int_{S^{n-1}} \log h_t(v) d\mu(v) \\ &= \frac{q\mu(S^{n-1})}{n(q+n)} \left[\log \frac{2\mu(S^{n-1})}{q+n} - \frac{n}{q} \log \Phi_q(K_t) \right] + \int_{S^{n-1}} \log h_{K_0} + t f d\mu. \end{aligned}$$

Therefore, by Theorem 3.10

$$\begin{aligned} 0 &= \frac{d}{dt} \Big|_{t=0} \int_{S^{n-1}} \log(\gamma(t) h_t(v)) d\mu(v) \\ &= -\frac{\mu(S^{n-1})}{n+q} \frac{\frac{d}{dt} \Big|_{t=0} \Phi_q(K_t)}{\Phi_q(K_0)} + \int_{S^{n-1}} f d\mu \\ &= -\frac{2\mu(S^{n-1})}{n+q} \Phi_q^{-\frac{n}{q}}(K_0) \int_{S^{n-1}} f(v) dF_{0,q}^{(a)}(K_0, v) + \int_{S^{n-1}} f d\mu \\ &= -\int_{S^{n-1}} f(v) dF_{0,q}^{(a)}(K_0, v) + \int_{S^{n-1}} f d\mu \end{aligned}$$

Since f is arbitrary, this implies the desired result. $\square$

Lemma 5.5. *Let $0 \neq q > -1$ and μ be a finite even Borel measure on S^{n-1} . If μ satisfies the strict subspace concentration condition; that is*

$$\frac{\mu(\xi \cap S^{n-1})}{\mu(S^{n-1})} < \frac{\dim \xi}{n},$$

for every proper subspace $\xi \subset \mathbb{R}^n$, then there exists an origin-symmetric convex body K_0 such that $\Phi_q(K_0) = \left[\frac{2\mu(S^{n-1})}{q+n} \right]^{\frac{q}{n}}$ and K_0 is a minimizer to (5.9).

Proof. Let K_i be a sequence of origin-symmetric convex body with $\Phi_q(K_i) = \left[\frac{2\mu(S^{n-1})}{q+n} \right]^{\frac{q}{n}}$ such that it is a minimizing sequence to (5.9). Let

$$E_i = \left\{ x \in \mathbb{R}^n : \frac{|x \cdot e_{1,i}|^2}{h_{1,i}^2} + \cdots + \frac{|x \cdot e_{n,i}|^2}{h_{n,i}^2} \leq 1 \right\}$$

be the John ellipsoid of K_i with $0 < h_{1,i} \leq h_{2,i} \leq \cdots \leq h_{n,i}$ and $e_{1,i}, \dots, e_{n,i}$ an orthonormal basis. Define the cross-polytope C_i to be the convex hull of $\pm h_{1,i}e_{1,i}, \dots, \pm h_{n,i}e_{n,i}$. John's theorem, together with the definition of C_i , implies that $C_i \subset K_i \subset nC_i$.

By the fact that $nC_i \supset K_i$ and Lemma 5.2, there exists $c_0 > 0$ such that

$$V(C_i) \geq n^{-n}V(K_i) \geq c_0 n^{-n} \Phi_q^{\frac{n}{n+q}}(K_i) = c_0 n^{-n} \left(\frac{2\mu(S^{n-1})}{q+n} \right)^{\frac{q}{n+q}} := c_1 > 0.$$

The volume formula for cross-polytopes now implies the existence of $c_2 > 0$ such that

$$h_{1,i}h_{2,i} \cdots h_{n,i} \geq c_2.$$

If the numbers $h_{1,i}, h_{2,i}, \dots, h_{n,i}$ were not uniformly (in i) bounded from above, then [8, Lemma 6.2] would imply the integral

$$\int_{S^{n-1}} \log h_{C_i} d\mu$$

is not bounded from above (as a sequence in i), which when combined with the fact that $C_i \subset K_i$, would imply the integral

$$\int_{S^{n-1}} \log h_{K_i} d\mu$$

is not bounded from above (as a sequence in i). This is a contradiction to K_i being a minimizing sequence. Hence, the sequence of convex bodies K_i is uniformly bounded. By Blaschke's selection theorem, there exists a subsequence K_{i_j} that converges in the Hausdorff metric to a compact convex set K_0 in $\mathbb{R}^n$. Argue as in Lemma 5.3 and we conclude that $K_0 \in \mathcal{K}_e^n$. Continuity of $\Phi_q(\cdot)$ on $\mathcal{K}_e^n$ now implies $\Phi_q(K_0) = \left[\frac{2\mu(S^{n-1})}{q+n} \right]^{\frac{q}{n}}$. Hence, K_0 is a minimizer. $\square$

Proof of Theorem 1.3. It follows directly from Lemmas 5.4 and 5.5. $\square$

At this point, we remark that although the estimates in Lemma 5.2 are two-sided, when proving Theorem 1.3, we only utilized the upper bound. However, the lower bound implies that the bound

$$\frac{1}{n+q} \log \Phi_q(K_i) \geq \frac{1}{n} \sum_{j=1}^n \log h_{j,i} + c \quad (5.10)$$

for some constant $c \in \mathbb{R}$, is the best possible; that is, it is not possible to replace the coefficient $\frac{1}{n}$ on the right-side of (5.10) by other coefficients in order to achieve better estimates. The coefficient $1/n$ appears in the subspace concentration condition (1.7).

6. MEASURE EQUATIONS RELATED TO AFFINE ISOPERIMETRIC INEQUALITIES.

Let $-1 < p \neq 0$. Recall that a similar argument to Lemma 5.2 shows that the $\text{GL}(n)$ -invariant ratio

$$\frac{V(R_p K)}{V(K)} \quad (6.1)$$

attains both (finite and positive) upper and lower bounds on $\mathcal{K}^n$. While a simple application of Jensen's inequality shows that it has an upper bound of 1 when $p < n$ and a lower bound of 1 when $p > n$, neither bound is sharp, as the equality condition of Jensen's inequality is never satisfied in this case. More generally, one considers the upper and lower bounds of the ratio

$$\frac{V(R_p K)}{V(R_q K)} \quad (6.2)$$

on $\mathcal{K}^n$ when nonzero p and q satisfies $-1 < p < q$. Equation (6.1) is a special case of (6.2) when either p or q is equal to n . It was shown by Gardner-Zhang [24, Corollary 5.6] that the lower bound of (6.2) is attained if and only if K is a simplex. An inequality of Berwald is a crucial ingredient of the proof. When p and q take special values, the Gardner-Zhang inequality (that is, the sharp upper bound of (6.2) along with its equality characterization) recovers the Rogers-Shephard inequality [59, 60] and the Zhang projection inequality [68].

Curiously, it is the upper bound of (6.2)—the isoperimetric direction, conjectured to be attained if and only if K is an ellipsoid—that turned out to be the more challenging direction of the two. Only very recently, this conjecture was shown to be true by Haddad-Ludwig [33, Theorem 19] in the special case when either p or q is equal to n . (In other words, when either the numerator or the denominator is $V(K)$.) The result of Haddad-Ludwig has been stated in the Introduction. See Theorem 1.4.

Remark 6.1. *The careful reader will note that Theorem 1.4 is valid for $p = 0$ —something we have not defined in the current paper. For its definition, see [24, Page 509].*

The characterization of the extremal bodies for the upper bound of (6.2) in the general case remains open. Armed with the tool of Theorem 1.1, we derive an alternate equation that the extremal bodies for the ratio (6.2) must satisfy.

Towards this end, we rewrite the ratio (6.2). By (2.6),

$$\frac{V(R_p K)}{V(R_q K)} = \tau_{p,q} V(K)^{-n(1/p-1/q)} \frac{\Phi_p^{\frac{n}{p}}(K)}{\Phi_q^{\frac{n}{q}}(K)},$$

where $\tau_{p,q} = \frac{(q+1)^{n/q}}{(p+1)^{n/p}}$. It is easier to work with the logarithm of the ratio,

$$\log \frac{V(R_p K)}{V(R_q K)} = \log \tau_{p,q} - n \left(\frac{1}{p} - \frac{1}{q} \right) \log V(K) + \frac{n}{p} \log \Phi_p(K) - \frac{n}{q} \log \Phi_q(K).$$

Note that a convex body K_0 is an extremal body for $\frac{V(R_p K)}{V(R_q K)}$ if and only if it is an extremal body for

$$J_{p,q}(K) := -\left(\frac{1}{p} - \frac{1}{q}\right) \log V(K) + \frac{1}{p} \log \Phi_p(K) - \frac{1}{q} \log \Phi_q(K).$$

Theorem 6.2. *Let p, q be nonzero distinct parameters that satisfy $p, q > -1$. If $K_0 \in \mathcal{K}^n$ is an extremal body for the ratio (6.2), then*

$$\left(\frac{1}{p} - \frac{1}{q}\right) \frac{1}{V(K_0)} S_{n-1}(K_0, \omega) + \frac{2}{q \Phi_q^{\frac{n}{q}}(K_0)} F_q^{(a)}(K_0, \omega) = \frac{2}{p \Phi_p^{\frac{n}{p}}(K_0)} F_p^{(a)}(K_0, \omega) \quad (6.3)$$

for every Borel subset $\omega \subset S^{n-1}$.

Proof. Note that K_0 is an extremal body for $J_{p,q}$. Let $f \in C(S^{n-1})$ be arbitrary and consider

$$K_t = [h_{K_0} + tf].$$

For sufficiently small $|t|$, the Wulff shape K_t defines a convex body in $\mathbb{R}^n$. Since $J_{p,q}(K_t)$ attains extremal value at $t = 0$, we have, by Theorem 1.1, that

$$\begin{aligned} 0 &= \left. \frac{d}{dt} \right|_{t=0} J_{p,q}(K_t) \\ &= -\left(\frac{1}{p} - \frac{1}{q}\right) \frac{1}{V(K_0)} \left. \frac{d}{dt} \right|_{t=0} V(K_t) + \frac{1}{p \Phi_p(K_0)} \left. \frac{d}{dt} \right|_{t=0} \Phi_p(K_t) - \frac{1}{q \Phi_q(K_0)} \left. \frac{d}{dt} \right|_{t=0} \Phi_q(K_t) \\ &= -\left(\frac{1}{p} - \frac{1}{q}\right) \frac{1}{V(K_0)} \int_{S^{n-1}} f(v) dS_{n-1}(K_0, v) + \frac{2}{p \Phi_p^{\frac{n}{p}}(K_0)} \int_{S^{n-1}} f(v) dF_p^{(a)}(K_0, v) \\ &\quad - \frac{2}{q \Phi_q^{\frac{n}{q}}(K_0)} \int_{S^{n-1}} f(v) dF_q^{(a)}(K_0, v). \end{aligned}$$

Since f is arbitrary, this implies the desired consequence. $\square$

Since $V(R_n K) = V(K)$, we immediately have the following corollary.

Corollary 6.3. *Let q be a nonzero parameter that satisfies $n \neq q > -1$. If $K_0 \in \mathcal{K}^n$ is an extremal body for the ratio (6.2), then*

$$\frac{n+q}{V(K_0)} S_{n-1}(K_0, \omega) = \frac{2n}{\Phi_q^{\frac{n}{q}}(K_0)} F_q^{(a)}(K_0, \omega), \quad (6.4)$$

for every Borel subset $\omega \subset S^{n-1}$.

Note that (6.4) implies that the measures $S_{n-1}(K_0, \cdot)$ and $F_q^{(a)}(K_0, \cdot)$ are proportional to each other. By Gardner-Zhang [24, Corollary 5.6] and Haddad-Ludwig [33, Theorem 19], ellipsoids and simplices have this property. A quick examination also suggests that, in fact, all *regular* polytopes and their $SL(n)$ images have this property—note that balls can be viewed as regular polytopes with an infinite number of facets.

Open problem. Are regular polytopes, balls, and their $SL(n)$ images the only convex bodies whose surface area measure and p -th fractional affine area measure are proportional to each other?

A positive answer to this will greatly reduce the number of possible extremizers to the ratio (6.1) and thus will potentially yield an alternative proof of both the Gardner-Zhang inequality [24, Corollary 5.6] and the Haddad-Ludwig inequality [33, Theorem 19].

When K_0 has C^2 boundary with everywhere positive Gauss curvature, (6.4) takes an even simpler form:

$$\tilde{V}_q(K_0, S_q K_0, z) = \frac{(n+q) \Phi_q^{\frac{n}{q}}(K_0)}{2n(q+1)V(K_0)},$$

for all $z \in \partial K$.

Open problem. Are ellipsoids the only convex bodies with C^2 boundary and everywhere positive Gauss curvature that satisfy

$$\tilde{V}_q(K_0, S_q K_0, z) \equiv \text{const},$$

for $z \in \partial K_0$?

Note that in this framework, to deal with the unsolved cases for the extremal body characterization of the ratio (6.2), one needs to characterize those convex bodies that satisfy (6.3).

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